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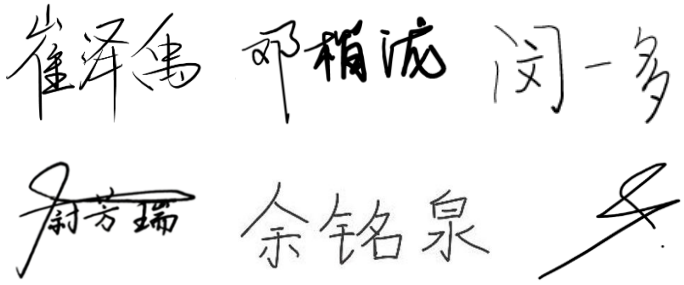
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Washington's Plug-in Vehicle Population, 2010–2026

DSC201 Statistical Programming Using R Assignment

Abstract

Washington State's plug-in vehicle population is recorded only as administrative transactions, not as a vehicle register. Using 1,926,507 title and registration records for 401,956 vehicles (January 2010 to September 2026), this report reconstructs the population at the vehicle-year level and asks how it has grown, aged and spread, and which vehicles remain. After cleaning (1,909,074 rows; 397,023 vehicles; 1,349,135 vehicle-years), the stock-and-flow identity reconciles exactly in every year and 66 of 70 independent benchmark figures match exactly or to rounding, with the four differences explained. The active fleet grew from 52,710 vehicles in 2019 to 270,098 in 2025 (5.1 times); mean age rose from 2.58 to 3.05 years; King County's share of vehicles fell 5.5 percentage points; and about 7–7.5% of active vehicles leave the records each year (no record in the following two years). After adjusting for age, no significant difference was detected between battery electric vehicles (BEVs) and plug-in hybrids (PHEVs) in leaving (odds ratio 1.03, 95% CI 0.97–1.10, $p = 0.26$). In the 2021 baseline, leaving is associated with prior transfers, use class and region, but is hard to predict (test AUC 0.57), because the reasons for leaving are not recorded. A stock-and-flow projection of the gap-filled count (286,478 in 2025) places the 2028 fleet at about 444,000 vehicles (95% interval 331,000–714,000). All results are associations rather than causal effects, and exits are observable only to 2023.

Contribution Table

Student ID	Name	Contribution
DSC2409007	Deng Kailong	Team leader; attended the dataset selection and planning meeting; prepared the presentation slides.
DSC2409071	Mounir Ben Hassine	Selected the dataset and planned the data processing and analysis workflow; implemented the analysis and visualizations.
DSC2409006	Cui Zeyu	Improved the analysis and visualizations; wrote the report draft and drew the figures.
DSC2409026	Min Yiduo	Improved the report; refined the analysis and visualizations.
DSC2409042	Yu Fangrui	Checked the analysis and report; revised the final report.
DSC2409043	Yu Mingquan	Checked the analysis and report; revised the final report.

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1. Context and Data Understanding

1.1 The dataset and its context

The analysis uses the Washington State Department of Licensing's *Electric Vehicle Title and Registration Activity* extract, published on data.wa.gov under the Open Database License. The extract was downloaded on 7 November 2026 (the source was last updated on 14 October 2026) and frozen by its SHA-256 hash, so every number in this report can be reproduced from the same file. It contains 1,926,507 rows and 33 columns describing 401,956 distinct vehicles between January 2010 and September 2026. Each row is one administrative transaction—an original title, a transfer title, an original registration, a registration renewal or a registration at transfer—rather than one vehicle. The dataset is therefore a transaction log, not a survey, and the population of vehicles has to be rebuilt from it.

The fields fall into four groups, summarised in Table 1. Vehicle fields describe the car itself (VIN prefix, make, model, model year, BEV or PHEV type and electric range); transaction fields describe the event (type, date, sale price, sale date, odometer and the New/Used label); location fields describe where the vehicle was registered (county, city, postal code, 2020 census tract, legislative district and electric utility); and policy fields record participation in the 2019 HB 2042 sales-tax exemption and the electric vehicle fees paid. Three measurement characteristics matter for the analysis. First, sale prices exist only on title rows, so price statistics use a subset of the records. Second, a new car's registration row is labelled "Used" even when the car is new, so entry routes must be taken from title rows rather than from the label. Third, electric range is almost never recorded for BEVs of model year 2021 or later, which limits the range analysis to older vehicles. County population estimates from the US Census Bureau and the Consumer Price Index from the Bureau of Labor Statistics are used as supporting data for per-resident rates and real prices.

Table 1: Feature groups in the source extract

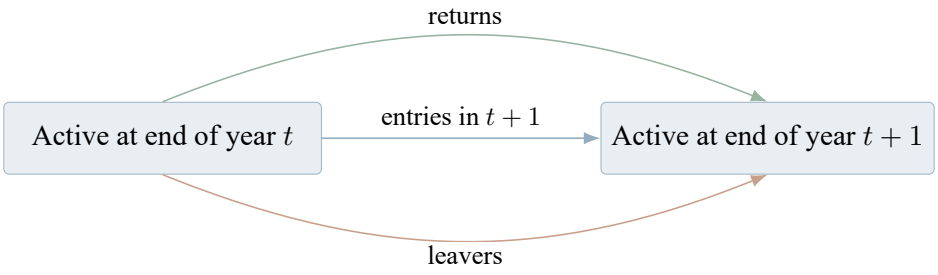
Group	Fields
Vehicle	VIN prefix, make, model, model year, BEV/PHEV type, electric range
Transaction	Transaction type, transaction date, sale price, sale date, odometer, New/Used label
Location	County, city, postal code, 2020 census tract, legislative district, electric utility
Policy	2019 HB 2042 tax-exemption eligibility, electric vehicle fees paid

Source: data dictionary in `outputs/tables/02_data_dictionary.csv` *(section 2.1 of the R script)*

The import, audit and cleaning steps are in sections 1–3 of the R script, and the full profile of every column is in section 2.1.

1.2 Indicators that summarise the population

Because the source records transactions rather than vehicles, the unit of analysis is the vehicle-year: a vehicle is *active* in a year if it has at least one record in that year. The population can also be counted with a one-year gap filled, which treats a vehicle as active when it has records in the years either side of a missing year; this gap-filled count is used wherever a single missing record would otherwise look like an exit. Four flows move the population between years: *entries* (the first Washington record of a vehicle), *returns* (a vehicle that reappears after a two-year absence), *leavers* (a vehicle with no record in the following two years) and county moves. Two further indicators summarise composition: vehicle age (year minus model year, floored at zero) and the BEV share of active vehicles. Figure 1 shows the accounting identity that links the stock to these flows, and Figure 2 shows the resulting stock over time.



$$\text{Active}_{t+1} = \text{Active}_t + \text{entries}_{t+1} + \text{returns}_{t+1} - \text{leavers}_t$$

leavers: no record in $t+1$ or $t+2$; returns: first record after a two-year absence

Figure 1: Stock-and-flow identity used for every year, on the gap-filled count. The identity is checked exactly in section 9.1 of the R script.

The fleet grew five-fold in six years, from 52,710 active vehicles in 2019 to 270,098 in 2025 (54,733 and 286,478 on the gap-filled count). Entries rose from 15,554 in 2019 to 60,788 in 2023 and stayed close to 72,000 a year in 2024 and 2025, while the BEV share rose from 69.2% to 79.8% and mean age from 2.58 to 3.05 years. The population is therefore becoming larger, older and more battery-electric at the same time. The 2024 leaver count is provisional and no 2025 count can be given yet: a vehicle is only classified as a leaver after two further years without a record, and the 2026 data end in September. The definitions and the flow table are produced in sections 4–6 of the R script, and Figure 2 in section 11.1.

1.3 Variability, distributions and correlation

The distribution of vehicle age is right-skewed in every year. In 2025 the mean age was 3.05 years and the median 2 years, with a standard deviation of 3.01 (variance 9.05), skewness of 1.32 and excess kurtosis of 1.30 (raw kurtosis 4.30): there are many young vehicles and a long tail of older ones. The mode (2 years) equals the median and lies below the mean, as expected for a right-skewed measure. In 2019 the mean was 2.58 years and the standard deviation 2.23. Real used prices, expressed in September 2026 dollars, are more strongly skewed: across 141,854

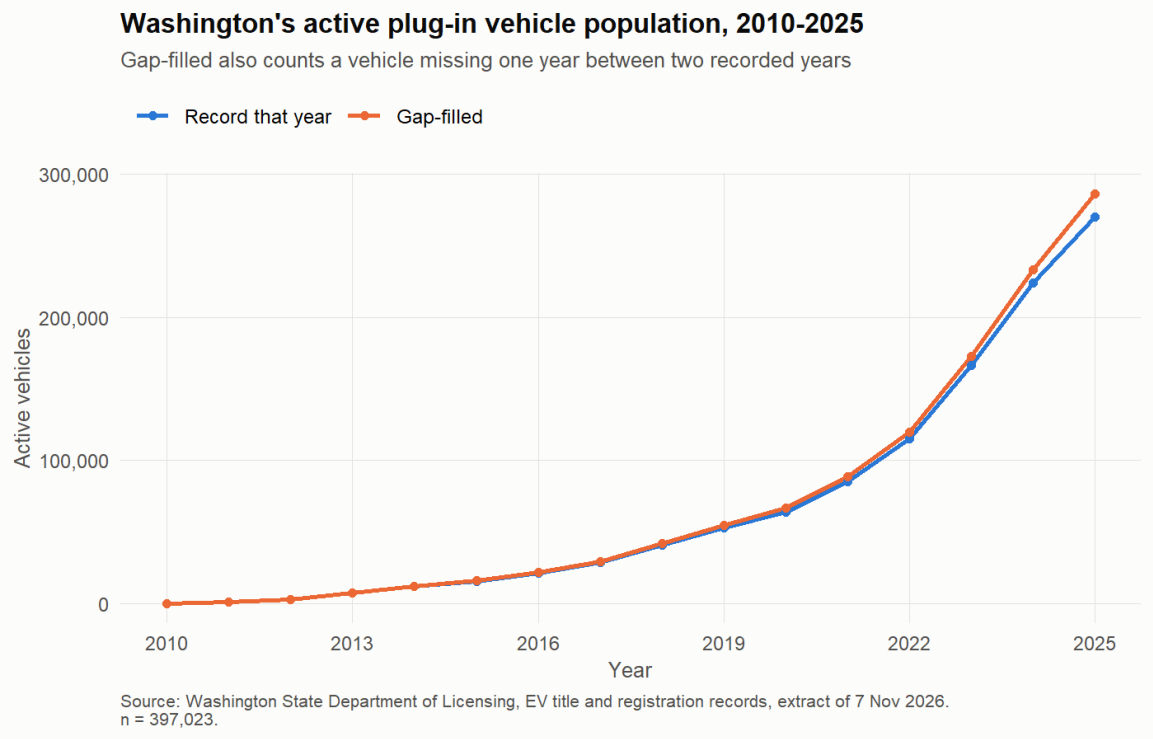


Figure 2: Active plug-in vehicles per year, counted directly and with a one-year gap filled

Table 2: Population stock and flows by year

Year	Active (simple)	Active (gap-filled)	Entries	Leavers after year	BEV %	Mean age
2010	80	80	80	4	97.5	3.79
2011	1,177	1,179	1,103	56	88.4	0.35
2012	2,932	2,958	1,835	104	62.9	0.50
2013	7,528	7,623	4,769	473	69.9	0.59
2014	11,821	12,125	4,973	942	69.4	1.04
2015	15,695	16,230	5,043	1,427	69.6	1.48
2016	20,973	21,716	6,895	1,830	66.0	1.89
2017	28,485	29,436	9,508	2,284	65.6	2.21
2018	40,746	42,131	14,913	3,046	66.2	2.26
2019	52,710	54,733	15,554	3,615	69.2	2.58
2020	63,724	66,708	15,492	4,360	71.7	2.92
2021	85,340	88,867	26,248	6,359	73.4	2.97
2022	114,914	119,640	36,674	7,940	76.2	2.90
2023	166,427	173,070	60,788	12,539	78.0	2.67
2024	224,133	233,487	72,036	20,215	79.1	2.76
2025	270,098	286,478	71,635	—	79.8	3.05
2026	242,352	242,352	49,477		80.6	3.55

Source: `outputs/tables/06_population_flows.csv`. *2026 covers January to September. The 2024 leaver count is provisional because 2026 is partial; a 2025 count needs records for 2026 and 2027.*

used sales the mean was $33,100, the median $28,099, the standard deviation $21,551, skewness 1.84 and raw kurtosis 10.25, so the median is a better summary of a typical sale than the mean. County stocks are more unequal still: in 2025 the mean county held 6,926 vehicles but the median only 1,185, with skewness 5.20, because King County alone held 134,145 vehicles. Table 3 collects the descriptive statistics, and Figures 3–6 show the corresponding distributions.

Table 3: Descriptive statistics for the main continuous measures

Measure	n	Mean	Median	Mode	SD	Skewness	Kurtosis
Vehicle age, 2019 (years)	52,710	2.58	2	1	2.23	0.79	3.46
Vehicle age, 2025 (years)	270,098	3.05	2	2	3.01	1.32	4.30
Used sale price, real ($)	141,854	33,100	28,099	26,000	21,551	1.84	10.25
Odometer at used sale (miles)	143,527	34,300	27,111	20,000	29,007	1.85	8.70
County stock, 2025 (vehicles)	39	6,926	1,185	—	22,001	5.20	30.27
County log growth, 2019–25	39	1.80	1.79	1.8	0.26	0.98	4.39

Source: `outputs/tables/10_descriptives.csv`. *The mode of a continuous measure uses rounded values (years, $1,000, 1,000 miles, log points); kurtosis is raw, so excess kurtosis is 3 less.*

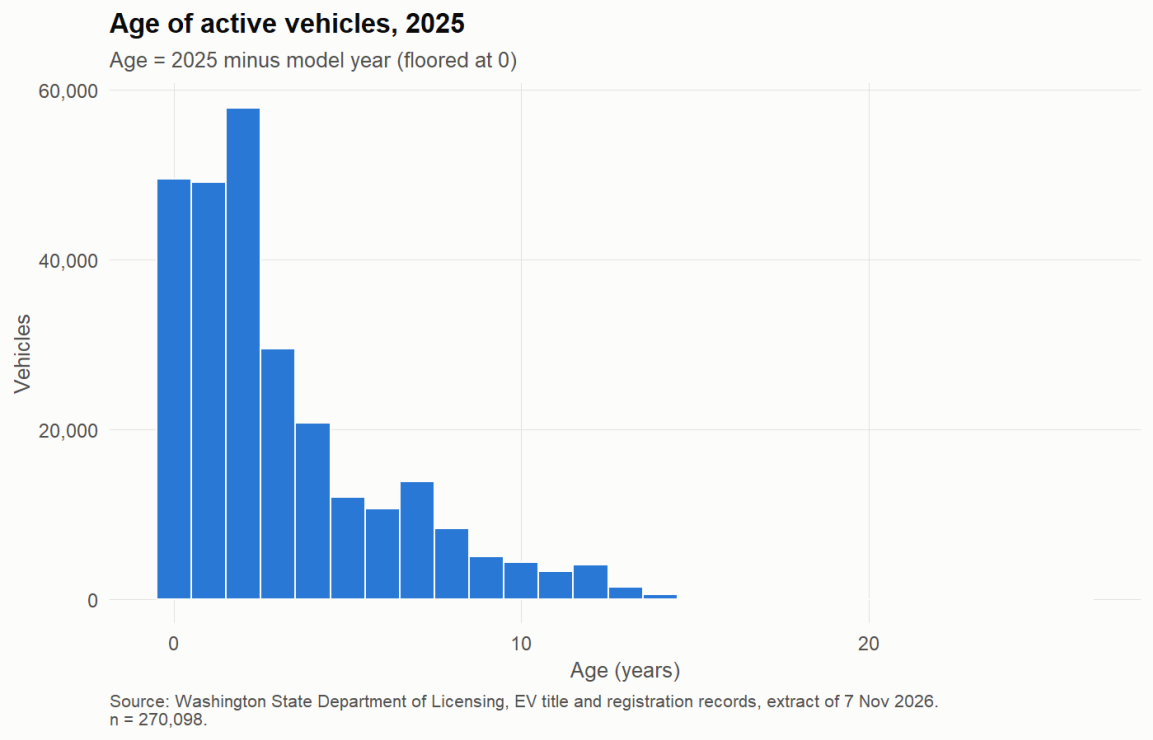


Figure 3: Distribution of vehicle age in 2025

Correlation between county measures is weak to moderate; the strongest pair is log 2019 stock and 2019 BEV share ($r = 0.56$). Growth from 2019 to 2025 is negatively associated with the log of 2019 stock (Pearson $r = -0.31$, $p = 0.058$; Spearman -0.33; Kendall -0.22), with covariance -0.169: smaller counties grew slightly faster. Growth is positively associated with the share of the 2019 stock that did not enter as new ($r = 0.42$, $p = 0.008$, covariance 0.010). Among used sales, the log real price falls with age ($r = -0.66$) and with odometer reading ($r = -0.49$), while age and odometer are themselves strongly related ($r = 0.69$); the two

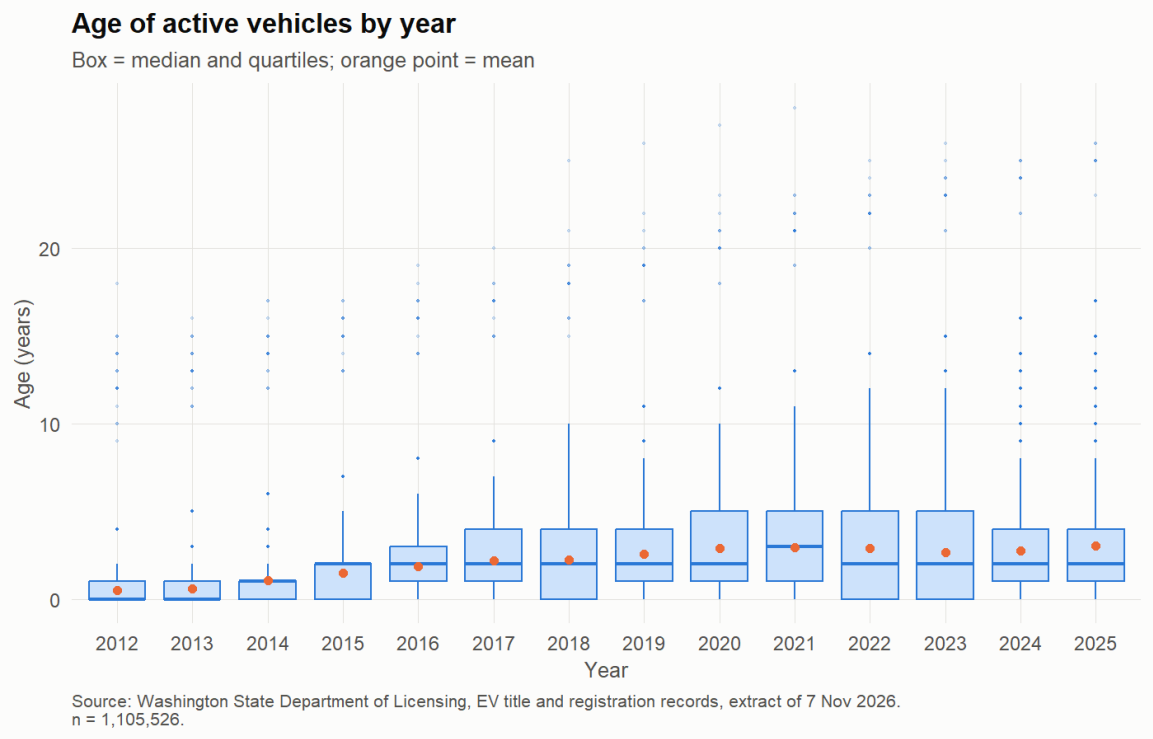


Figure 4: Vehicle age by year (boxplots; the orange point is the mean)

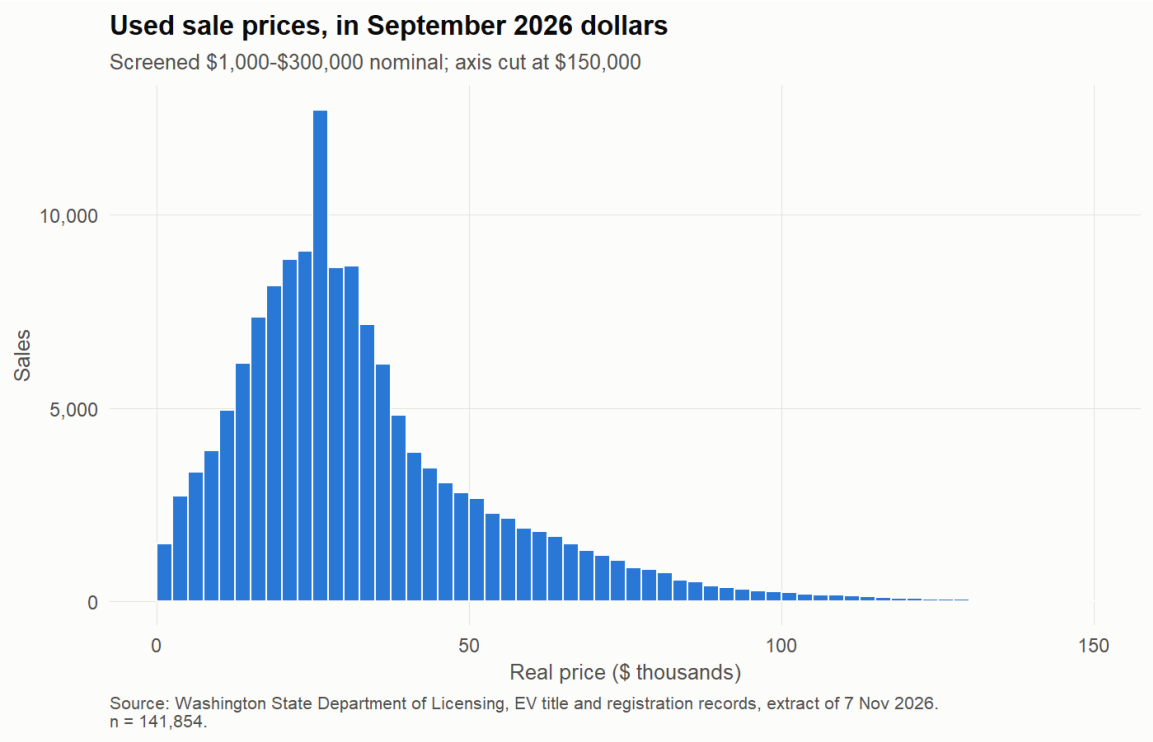


Figure 5: Distribution of real used sale prices

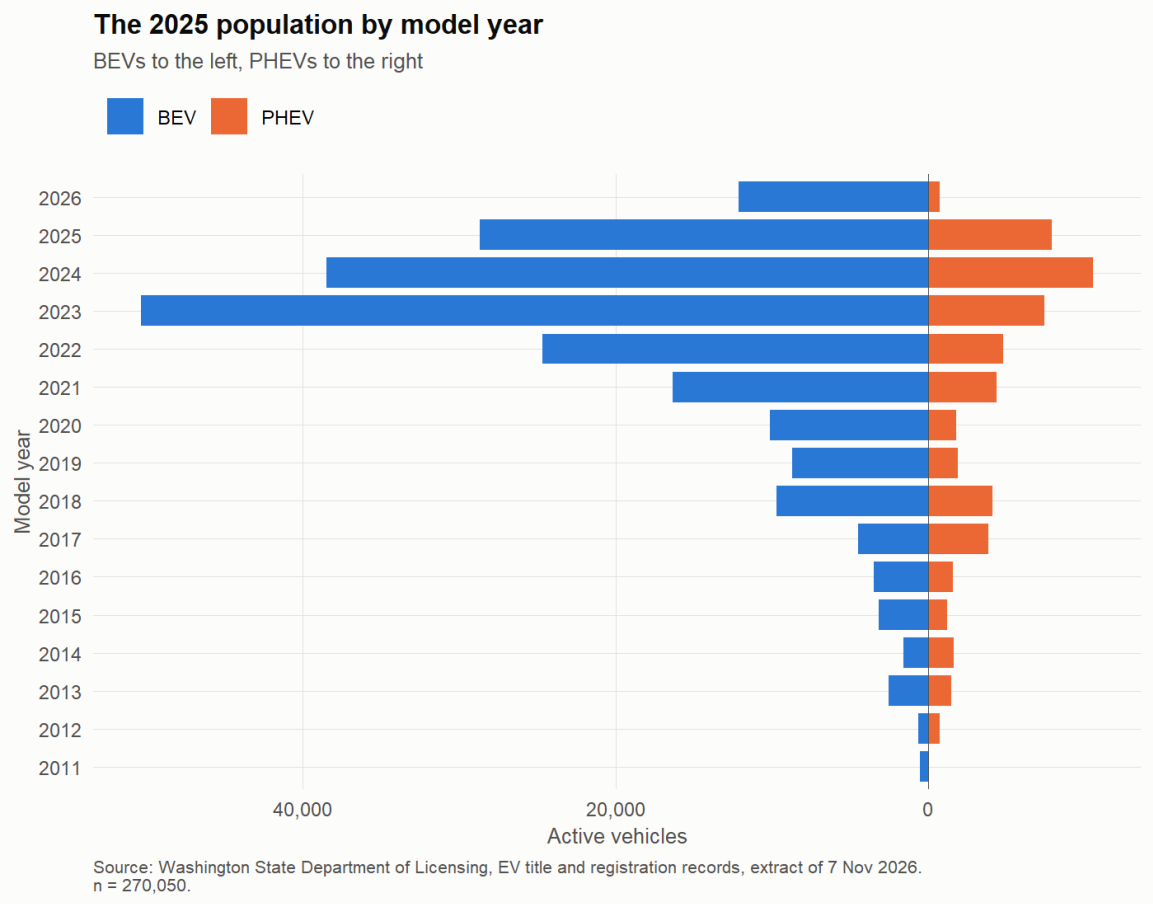


Figure 6: Active vehicles in 2025 by model year and type

therefore carry overlapping information about price. A useful way to read these measures is the standard score: King County sits 2.44 standard deviations above the mean on 2019 log stock but 1.04 below it on growth, which is why its share of the population can fall while its absolute stock rises. Figure 7 shows the county relationship directly. The Pearson and Spearman coefficients agree in sign for every pair. Correlation, however, describes association only. Table 3 and Figures 3–6 come from sections 10.4–10.5 and 11.7–11.12 of the R script, and the scatterplot in Figure 7 from section 11.11.

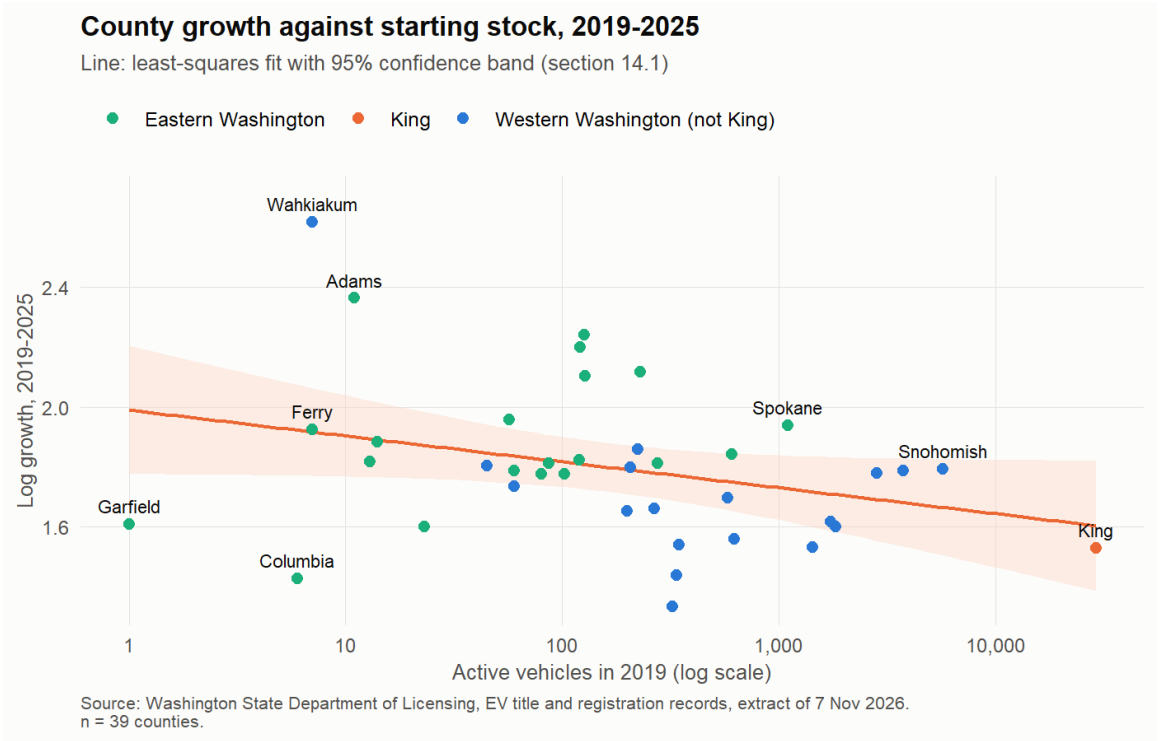


Figure 7: County log growth, 2019–2025, against log 2019 stock, with a least-squares line and 95% confidence band

1.4 Which vehicles, and where

The fleet is dominated by a small number of makes and by BEVs. In 2025 Tesla accounted for 111,984 active vehicles, more than six times the next make (Chevrolet, 18,341), and the ten largest makes covered 81.9% of the population (Figure 8). BEVs made up 79.8% of active vehicles and PHEVs 20.2% (Figure 9), continuing the shift towards battery-electric cars seen in Table 2. Table 4 gives the frequency table of active vehicles by year and type.

Geographically, the population remains concentrated but is spreading. King County's share of active vehicles fell from 55.1% in 2019 to 49.7% in 2025 (Figure 10), even though its absolute stock rose from 29,067 to 134,145 vehicles. The used share of title activity is lower in King County than in the rest of the state in every year shown (Figure 11), which is consistent with a larger share of new vehicles in the Seattle area. Figure 12 decomposes how vehicles first enter the Washington records: the great majority arrive as new titles, a smaller group as first

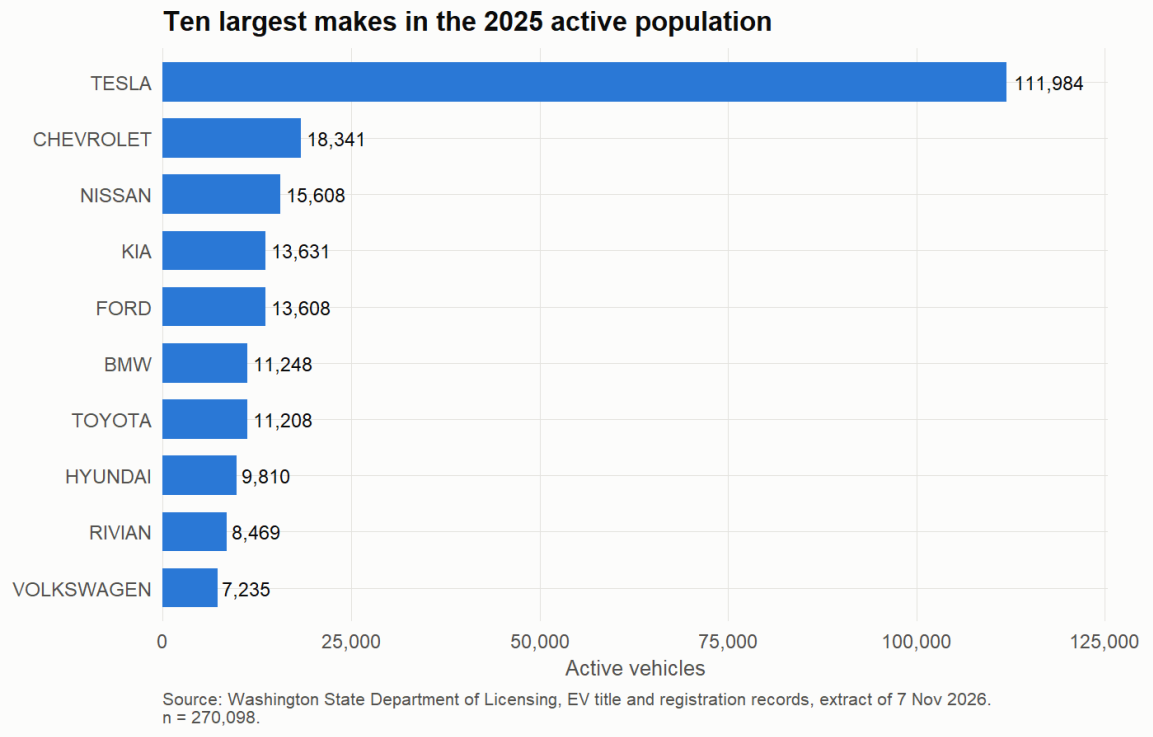


Figure 8: Ten largest makes among active vehicles in 2025

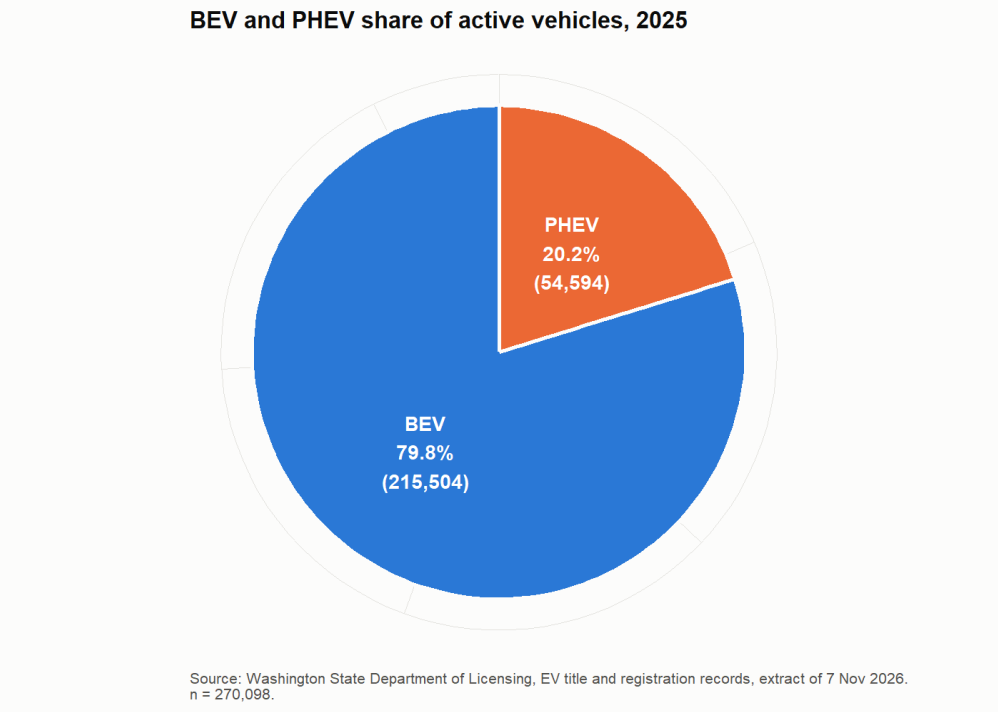


Figure 9: BEV and PHEV share of active vehicles in 2025

Washington titles for used cars, and a small group through registration only. Frequency tables for makes, counties and entry routes, together with the vehicle count by year and type, are written to `outputs/tables/10_freq_*.csv`; the figures are produced in sections 10.2 and 11.2–11.6 of the R script.

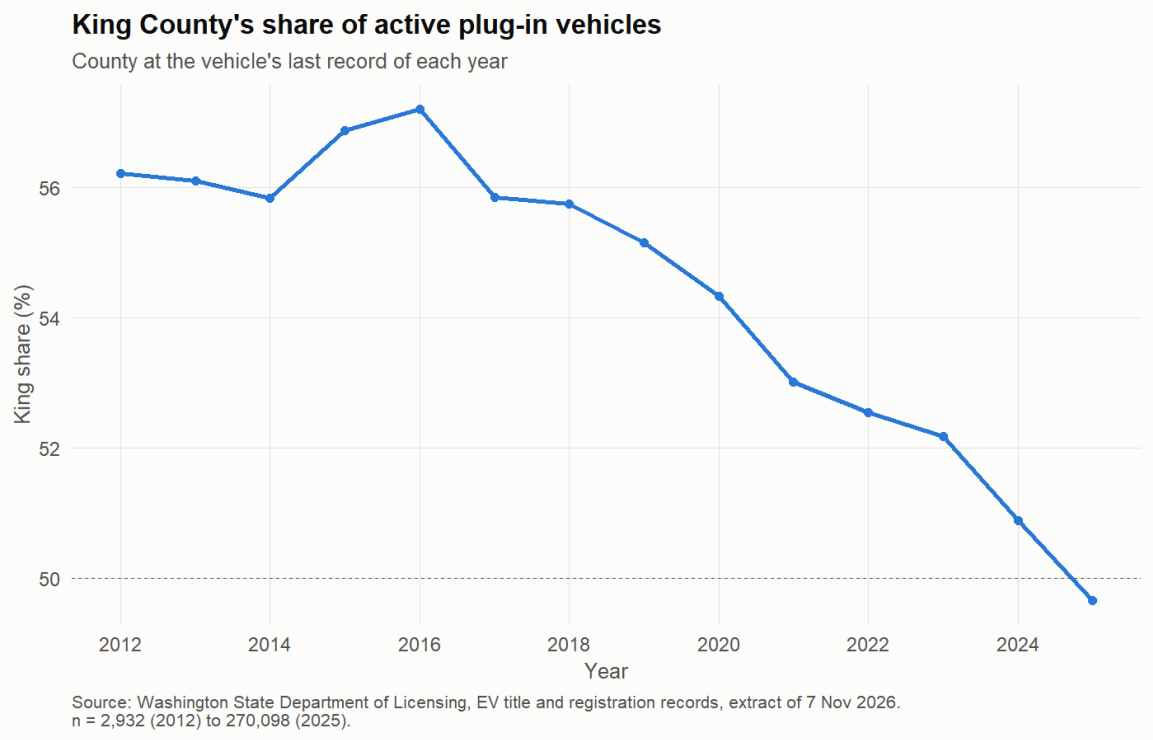


Figure 10: King County's share of active vehicles

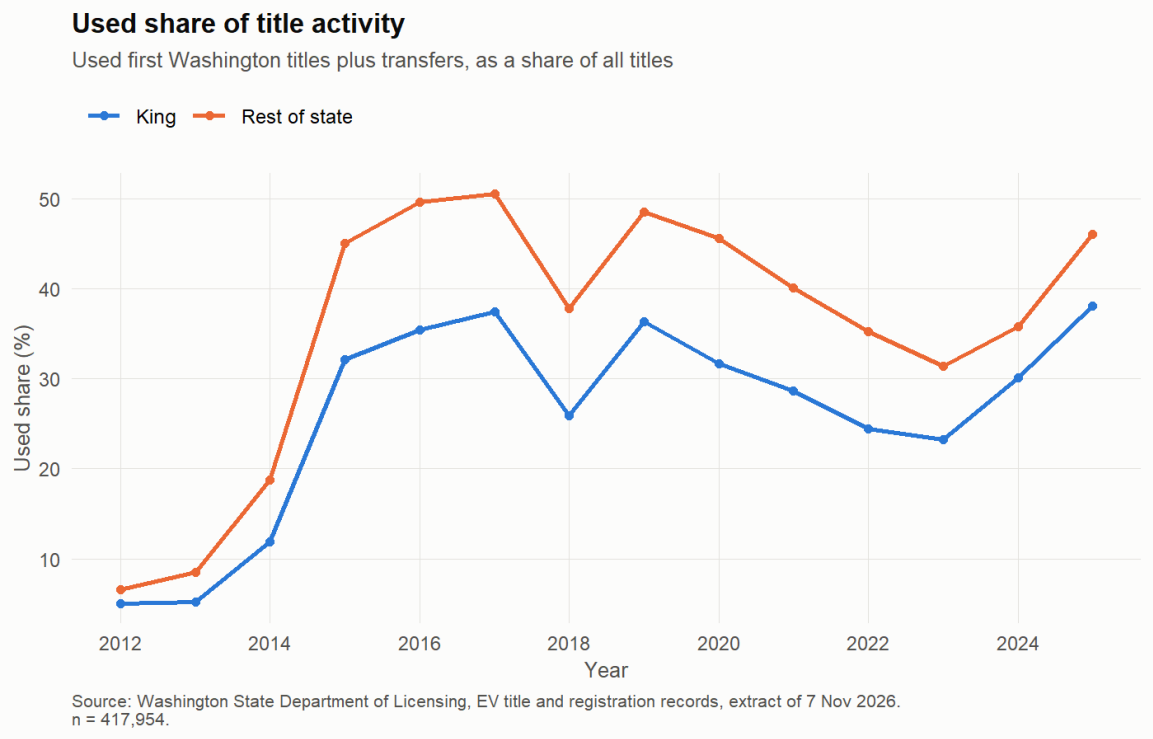


Figure 11: Used share of title activity, King County and the rest of the state

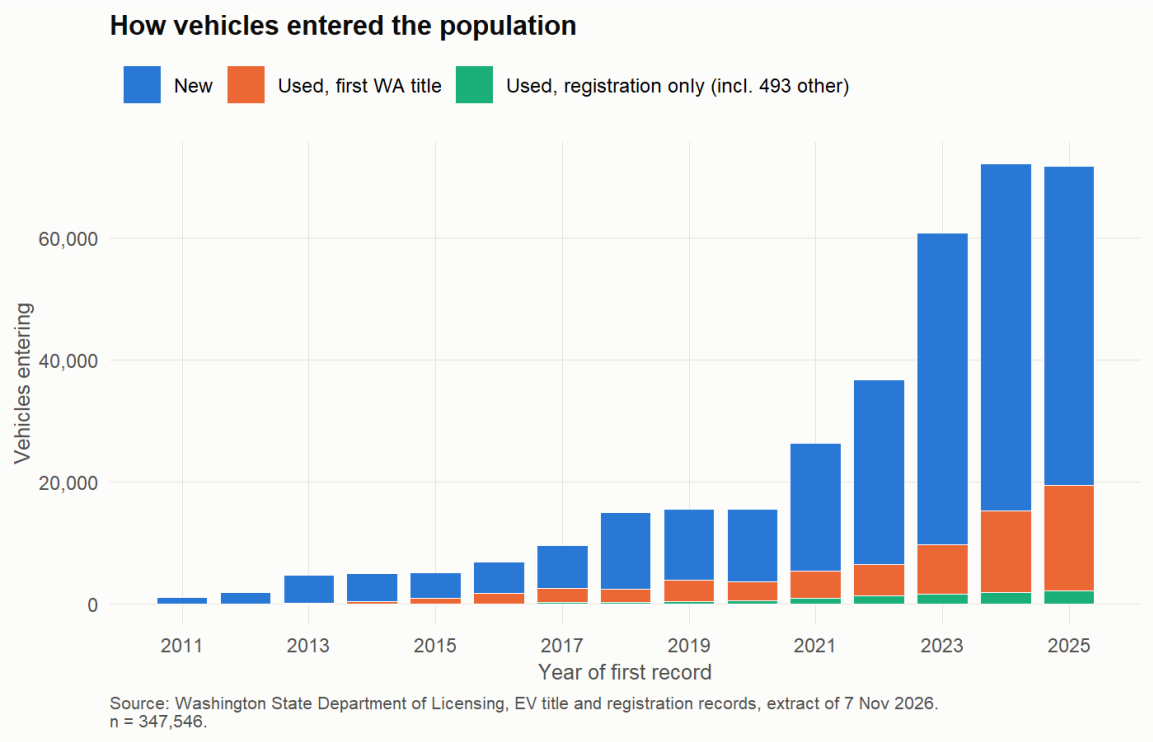


Figure 12: How vehicles first enter the Washington records, by year

Table 4: Active plug-in vehicles by year and type (frequency table)

Year	BEV	PHEV	Total	BEV %
2010	78	2	80	97.5
2011	1,041	136	1,177	88.4
2012	1,843	1,089	2,932	62.9
2013	5,261	2,267	7,528	69.9
2014	8,200	3,621	11,821	69.4
2015	10,927	4,768	15,695	69.6
2016	13,834	7,139	20,973	66.0
2017	18,696	9,789	28,485	65.6
2018	26,960	13,786	40,746	66.2
2019	36,478	16,232	52,710	69.2
2020	45,685	18,039	63,724	71.7
2021	62,603	22,737	85,340	73.4
2022	87,533	27,381	114,914	76.2
2023	129,893	36,534	166,427	78.0
2024	177,323	46,810	224,133	79.1
2025	215,504	54,594	270,098	79.8
2026 (Jan–Sep)	195,415	46,937	242,352	80.6

Source: `outputs/tables/10_freq_vehicles_by_year_type.csv`*, section 10.2 of the R script.*

2. Data Pre-processing

2.1 Import checks

The raw file is verified before it is used: its SHA-256 hash and row count must match the frozen extract, otherwise the script stops. Every column is read as text first, so that no value is changed silently on import, and is then parsed with explicit checks on dates, model year and price. The script also stops if any transaction date fails to parse or if the year field disagrees with the transaction date. Date parsing is forced to English month names, because otherwise every date becomes missing on a computer with a non-English locale, and no error is raised. Figure 13 summarises the path from the raw transactions to the analysis: cleaning, the vehicle-year and vehicle-history tables, the reconciliation checks, and the statistical methods.

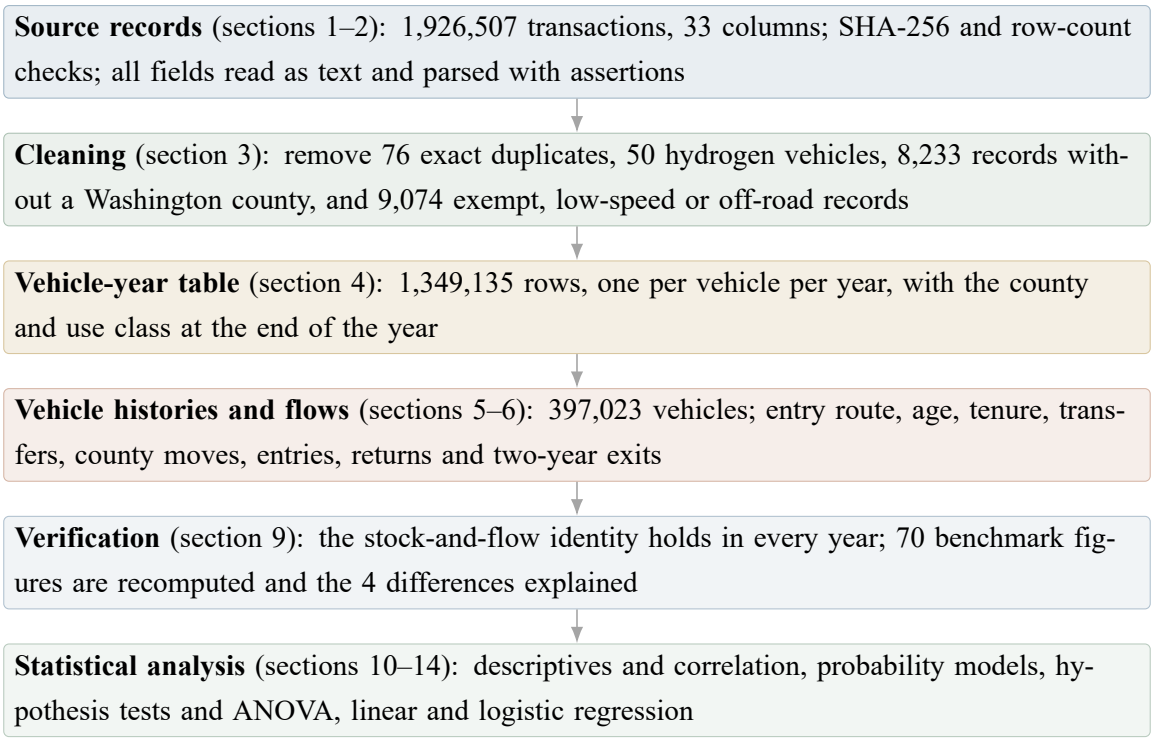


Figure 13: Data pipeline from transactions to the statistical analysis. Section numbers refer to the R script.

2.2 Cleaning rules, step by step

The cleaning rules address four problems that would otherwise distort the population. Exact duplicate rows (all 33 columns identical) are removed first. Hydrogen vehicles are removed because they are not plug-in vehicles. Records without a Washington address and county are removed, because the question concerns vehicles in Washington and every geographic measure needs a county. Exempt use classes are removed because they have no annual renewal, so leaving cannot be measured for them, and low-speed and off-road classes are removed because they are not general road vehicles. Table 5 reports the rows, vehicles and rows removed at each step; the study population is 1,909,074 rows and 397,023 vehicles.

The rules are also tested for how much they matter. Table 6 repeats the headline counts after each rule and, as a sensitivity check, on passenger-class vehicles only. Across the main rules the 2025 count changes by under 1% and King County's share by under 0.3 percentage points, so no conclusion depends on a particular cleaning decision.

2.3 Inconsistencies corrected

Several inconsistencies in the raw fields are corrected explicitly. The New/Used label is the most important: 305,781 of the 310,820 new-car titles have a same-day registration labelled "Used", because registration rows are always labelled Used. The entry route is therefore taken from the title row, not from the label (section 5.1). A vehicle can also appear in several counties within one year (13,677 vehicle-years); the county at the last record of the year is used, so county totals

Table 5: Records kept at each cleaning step

Step	Rows	Vehicles	Rows removed	Reason
Raw file	1,926,507	401,956		
Remove exact duplicate rows	1,926,431	401,956	76	
Remove hydrogen vehicles	1,926,381	401,935	50	
Keep Washington addresses with a county	1,918,148	401,256	8,233	
Remove exempt use classes	1,909,358	397,082	8,790	No annual renewal, so exit cannot be measured
Remove low-speed and off-road classes	1,909,074	397,023	284	Not general road vehicles
Vehicle-years	1,349,135	397,023		
Vehicle histories	397,023	397,023		

Source: `outputs/tables/09_flow_table.csv`. *The cleaning rules are applied in this order in section 3 of the R script.*

add up exactly to the state total (section 9.2). Fields that should never change for a vehicle—type, model year, make and model—were checked, and none changes (section 2.4).

Table 6: How much each cleaning rule changes the headline numbers

After rule	Vehicles ever	Active 2025	King share 2025 (%)
After removing duplicates	401,956	271,965	49.47
After removing hydrogen	401,935	271,957	49.47
After keeping Washington addresses	401,256	270,909	49.69
After removing exempt classes	397,082	270,142	49.66
After removing low-speed classes (study population)	397,023	270,098	49.67
Sensitivity: passenger class only	383,262	259,792	49.94

Source: `outputs/tables/09_rule_effects.csv`, *computed in section 9.3 of the R script.*

Prices require a separate cleaning path, summarised in Table 7. Sale prices exist only on title rows. Seventeen prices whose digits are the sale date (for example $62,720 on 27/06/2020) are removed, the range is kept to $1,000–$300,000, 23 vehicle-dates with conflicting same-day prices (46 rows) are set aside and 34 repeated rows of the same sale merged, and the remainder is converted to September 2026 dollars with the CPI-U. October 2025 has no CPI in the extract, so those sales drop out. Odometer readings are kept as a separate sample with their own checks, because a mileage value is only meaningful on a sale.

2.4 Feature engineering

The central feature-engineering step turns transactions into a population. A vehicle-year table gives one row per vehicle per year (1,349,135 rows) and a vehicle-history table gives one row per vehicle (397,023 rows), with the first and last year, the entry route, the number of prior transfers and any county moves. Two features require a rule rather than a simple sum. A vehicle is treated as leaving after a year if it has no record in the following two years, and a vehicle that reappears

Table 7: Price records kept at each step

Step	Rows
Title rows	483,815
Price recorded	428,759
Sale date recorded	428,759
Not date-like	428,742
$1,000 to $300,000	427,500
After removing same-day conflicts (sale events)	427,420
With CPI (excludes October 2025)	423,221
Used sale events	144,240
Used sale events in the mileage sample	143,527

Source: `outputs/tables/07_price_flow.csv`*, section 7 of the R script. The descriptive price statistics use 141,854 used sales with a usable real price; the specification model in Section 3.4 of this report uses 141,153 sales with complete age, mileage and make fields.*

after such a gap is counted as a return; the gap-filled count treats a one-year gap as still active. Figure 14 illustrates the four cases that these definitions separate. Other derived features are age, tenure in the records, region (King, western Washington outside King, or eastern Washington), county size class and real price.

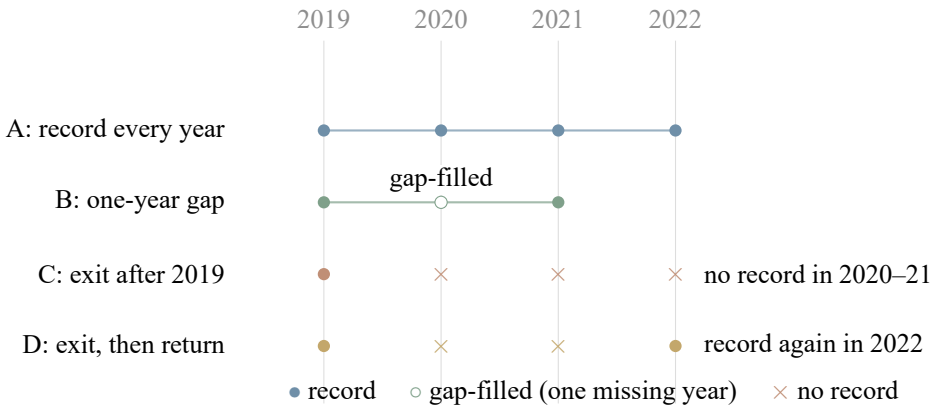


Figure 14: Entry, exit and return on a vehicle timeline. Two consecutive years without a record define an exit; a later record is a return. The definitions are implemented in section 6 of the R script.

Missing values and outliers are handled at this stage rather than dropped silently. Sale dates are blank on 73.5% of rows, legislative district on 0.43% and county on 0.01%; the county rule removes the latter, and district analyses are restricted to vehicles with a district. Electric range is almost never recorded for BEVs of model year 2021 or later, so the range analysis is limited to vehicles of model year 2020 or earlier. Prices are screened as described above; no age or price-tail outliers are removed, because the long right tail is genuine and is handled by medians, logs and robust standard errors, and the mileage sample excludes 54 readings that fell and 11 above 400,000 miles. Standardisation is used where a common scale helps interpretation: logarithms for stock, growth and price, real prices for inflation, and z-scores for county comparisons. The feature definitions are in sections 4–6 of the R script.

2.5 Verification

The population is checked in three independent ways before any statistical method is applied. First, the stock-and-flow identity of Figure 1 holds exactly for every year: the change in the gap-filled stock equals entries plus returns minus leavers. Second, 70 benchmark figures computed separately from the data portal were recomputed from the same extract; 66 match exactly or to rounding, and the four differences are explained (a difference of under 0.01% in two counts of cars by number of sales, the exclusion of October 2025 because it has no CPI, and the specification model's use of real prices and grouped makes). Third, ten vehicle histories were checked by hand across the main cases.

Two limitations follow from the data. Leaving is inferred from missing records, so the analysis cannot distinguish a vehicle that moved out of state from one that was taken off the road. The 2025 gap-filled count and the 2024 leaver count are provisional because 2026 is partial, covering only January to September. These limits are carried through to the conclusions. The reconciliation and benchmark checks are in section 9 of the R script.

3. Statistical Analysis and Modeling

3.1 Probability and distributions

Weekly new-vehicle entries are a discrete counting process with 873 weeks. A Poisson model is far too narrow: its 95% band covers only 37% of weeks, because the variance is much larger than the mean. A negative binomial model fits much better (AIC 9,571 against 39,696), its 95% band covers 95% of weeks, and there are 39 zero weeks, against 38 expected under the negative binomial and 45 under the Poisson (Figure 15). The negative binomial is therefore the better model for the arrival of new vehicles, although some overdispersion remains (Pearson dispersion 4.4).

The relationship between a discrete count and the normal distribution is shown by a binomial example. Of the 4,843 vehicles of model year 2013 that were active in 2021, 400 left by the end of 2022 (8.3%), against 6.9% expected from the 2019 two-year exit rate; the exact upper-tail probability is $P(X \geq 400) = 0.0001$. The normal approximation to the binomial improves as the number of trials grows: with a continuity correction its error is -0.083 at $n = 10$ and -0.007 at $n = 1{,}000$ (Figure 16).

The number of sale events per car, among cars with at least one priced sale, is close to a geometric distribution overall ($\hat{p} = 0.85$, goodness-of-fit $p = 0.21$), but not within single model years ($p < 0.001$): the overall fit is a mixture of cohorts with different selling behaviour (Figure 17). Finally, the sampling distribution of the mean is illustrated with vehicle age, which has skewness 1.32. 2,000 samples of size n produce a distribution whose standard deviation matches $\sigma/\sqrt{n}$ (0.133 against the theoretical 0.135 at $n = 500$), and whose skewness falls from 0.51 at $n = 5$

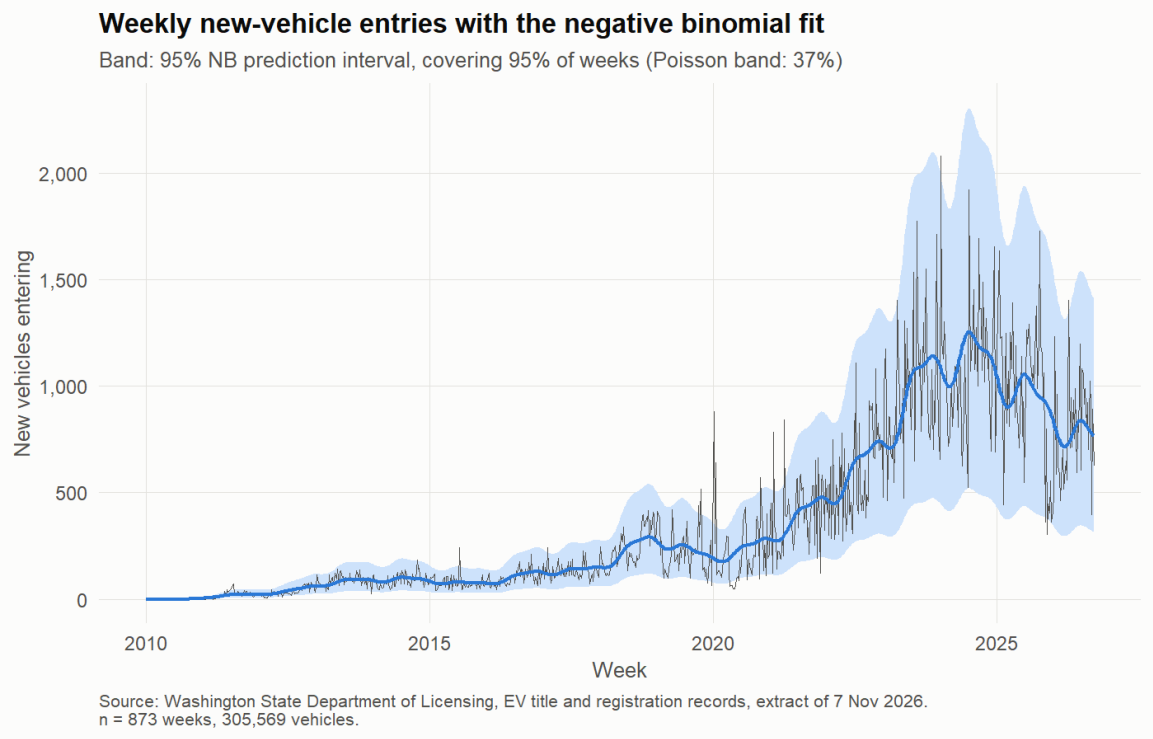


Figure 15: Weekly new-vehicle entries with the negative binomial fit and its 95% prediction band

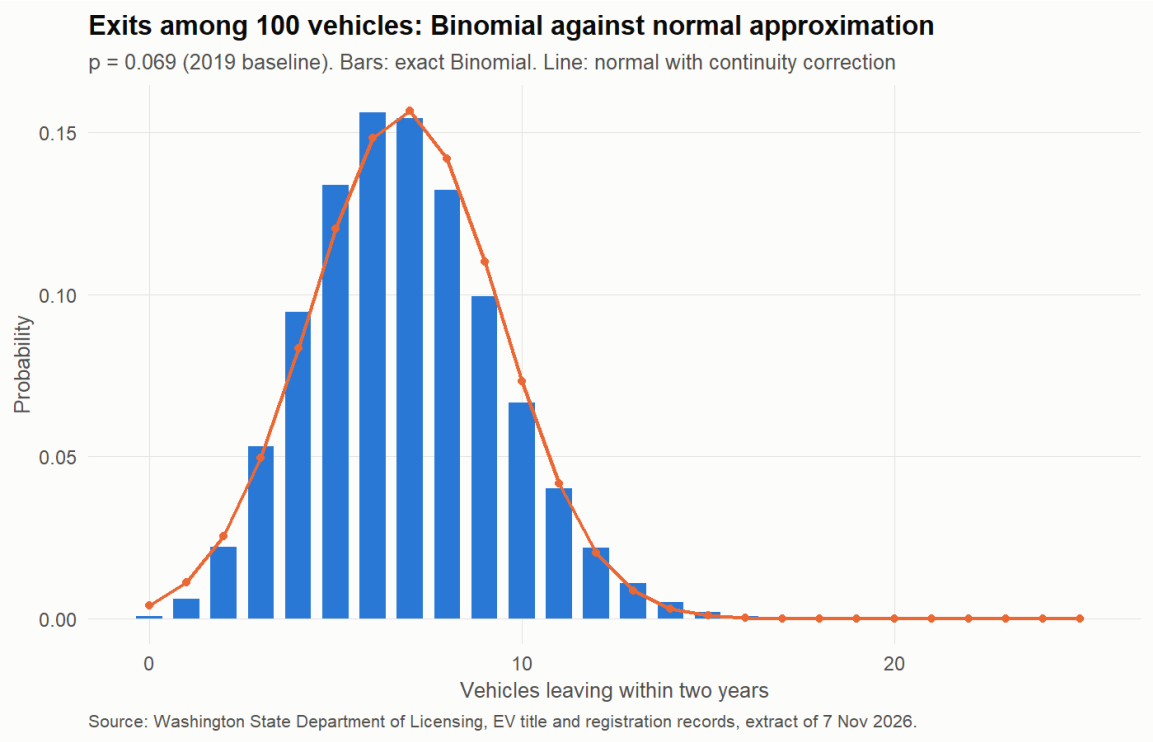


Figure 16: Binomial probabilities for exits among 100 vehicles ($p = 0.069$, the 2019 exit rate) and the normal approximation

to 0.02 at $n = 500$, as the central limit theorem predicts (Figure 18). The four models are implemented in sections 12.1–12.4 of the R script.

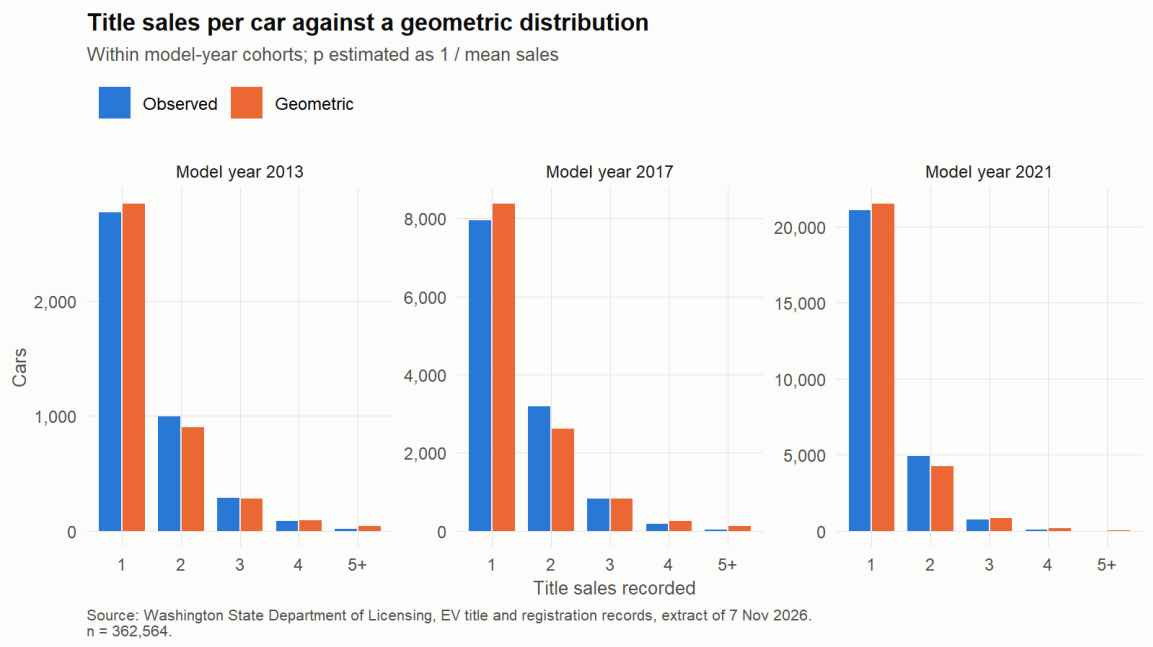


Figure 17: Sale events per car against a geometric model in three model-year cohorts

3.2 Hypothesis tests

The main questions are tested with two hypothesis tests, supported by a set of sensitivity and supporting tests. Table 8 reports the main and sensitivity results with effect sizes (adjusted exit probabilities for the logistic rows) and whether each test remains significant after a Bonferroni correction across the 9 non-sensitivity (main and supporting) tests in this section; Table 9 reports the supporting tests. All tests use $\alpha = 0.05$; the Bonferroni threshold is $0.05/9 \approx 0.0056$. Effect sizes are reported because with hundreds of thousands of observations a very small difference can be statistically significant.

The main spread test asks whether King County's share of active vehicles is the same in 2025 as in 2019. It is not: the two-proportion z statistic is -23.0, the share fell 5.48 percentage points (95% CI 5.02 to 5.94), and $p \approx 3 \times 10^{-117}$; the effect size is small (Cohen's $h = -0.110$) but the change is precisely estimated. One caveat matters for interpretation: many vehicles appear in both years, so the two samples are not independent, and the nominal standard error is too small. The gap-filled count gives the same estimate (-5.40 points), which is reassuring, but the test is best read as a description of a large compositional shift rather than as a test of two independent populations.

The main retention test asks whether, after adjusting for age, BEVs and PHEVs are equally likely to have no record in the next two years, using the 2021 baseline. The PHEV odds ratio is 1.035 (95% CI 0.975 to 1.098) with a likelihood-ratio $p = 0.264$, so the hypothesis of equal retention is not rejected; the adjusted exit probabilities are 7.39% for BEVs and 7.62% for PHEVs. The

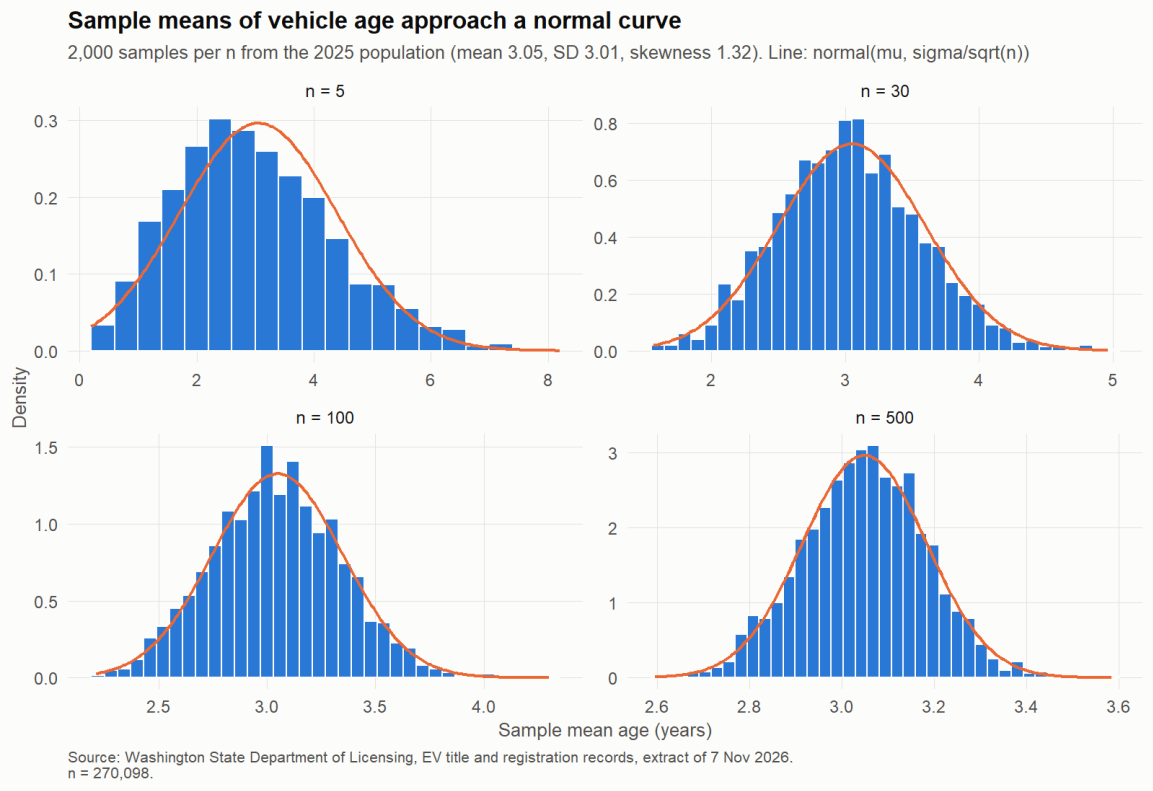


Figure 18: Sampling distribution of mean vehicle age for sample sizes from 5 to 500

Table 8: Main and sensitivity hypothesis tests

Test	Role	Estimate	95% CI	p	Effect size	Bonf.
Two-proportion z, King share 2025 vs 2019	Main: spread	-5.48	$-5.94, -5.02$	< 0.001	Cohen's $h = -0.110$	yes
Two-proportion z, King share, gap-filled	Sensitivity	-5.40	$-5.85, -4.94$	< 0.001	Cohen's $h = -0.108$	yes
Logistic LR, PHEV vs BEV exit, 2019 baseline	Sensitivity	0.897	0.832, 0.968	0.005	BEV 7.08%, PHEV 6.41%	yes
Logistic LR, PHEV vs BEV exit, 2021 baseline	Main: retention	1.035	0.975, 1.098	0.264	BEV 7.39%, PHEV 7.62%	no
Logistic LR, PHEV vs BEV exit, 2023 baseline	Sensitivity	0.967	0.924, 1.012	0.143	BEV 7.59%, PHEV 7.35%	no

Source: `outputs/tables/13_hypothesis_tests.csv`*, sections 13.1–13.2. Estimates are percentage points for the proportion tests and odds ratios for the logistic tests; the effect size for the logistic rows is the adjusted two-year exit probability by type. CI = confidence interval.*

2019 baseline shows a small but significant difference in the opposite direction (odds ratio 0.897, $p = 0.005$, so PHEVs were less likely to leave), while the 2023 baseline is again not significant (0.967, $p = 0.143$). The evidence therefore shows no significant difference between the two types in the recent baselines, with a difference in the 2019 and earlier baselines.

The supporting tests check the ingredients of that conclusion. A one-sample z test compares the mean age of a random sample of 500 vehicles from 2025 (3.12 years, 95% CI 2.86 to 3.39) with the 2019 mean of 2.58, using the 2025 population standard deviation of 3.01 ($z = 4.03$, $p < 0.001$, $d = 0.18$). A second one-sample z test compares the 2023 exit share (7.53%, 95% CI 7.41 to 7.66) with the 2019 rate (6.86%; $z = 10.91$, $p < 0.001$). A one-sample t test shows that the mean log growth of the 38 counties other than King (1.81) exceeds King's growth ($t = 6.56$, $p < 0.001$, $d = 1.06$). A paired t test shows that county BEV share rose 17.6 percentage points between 2019 and 2025 (95% CI 14.5 to 20.7, $t = 11.55$, $p < 0.001$), and a Welch t test shows that BEVs that left were 0.83 years younger at their last record than PHEVs that left ($p < 0.001$, $d = -0.31$). Finally, vehicles that left the 2021 cohort were 0.21 years older than those that stayed ($p < 0.001$) and exit rates differ by prior transfers (7.1%, 8.5% and 10.7% for 0, 1 and 2+ transfers; $\chi^2 = 84.8$, df 2, $p < 0.001$, Cramér's $V = 0.032$). These tests identify the features that separate leavers from stayers, and they motivate the logistic model in Section 3.4 of this report. The tests are implemented in sections 13.1–13.8 of the R script.

Table 9: Supporting hypothesis tests

Test	Role	Estimate	95% CI	p	Effect size	Bonf.
One-sample z (mean): age of a 2025 sample vs 2019 mean	Supporting	3.124	2.860, 3.388	< 0.001	$d = 0.18$	yes
One-sample z (proportion): 2023 exit share vs 2019 rate	Supporting	7.534	7.407, 7.661	< 0.001	$h = 0.026$	yes
One-sample t: county log growth vs King's growth	Supporting	1.806	1.720, 1.891	< 0.001	$d = 1.06$	yes
One-sample t: counties with 10+ vehicles in 2019	Sensitivity	1.795	1.718, 1.873	< 0.001	—	yes
Paired t: county BEV share 2025 vs 2019	Supporting	17.605	14.506, 20.704	< 0.001	$d_z = 1.95$	yes
Welch t: age at last record, BEV vs PHEV	Supporting	-0.829	-0.892, -0.766	< 0.001	$d = -0.31$	yes
Welch t: age at 2021 baseline, left vs stayed	Supporting	0.215	0.142, 0.288	< 0.001	$d = 0.08$	yes
Chi-square: exit by prior transfers (0, 1, 2+)	Supporting	84.8	—	< 0.001	$V = 0.032$	yes

Source: `outputs/tables/13_hypothesis_tests.csv`*, sections 13.3–13.8. Estimates are in the units of the tested quantity (years, percentage points, or the mean); the chi-square row reports the test statistic.*

3.3 Analysis of variance

The one-way analysis of variance asks whether county growth from 2019 to 2025 differs by county size class (small, medium, large: thirds of the 2019 stock, 13 counties each). The test is chosen by the assumption checks, not by the resulting p-value: Levene's test does not reject equal variances ($p = 0.559$) and the Shapiro–Wilk test does not reject normal residuals ($p = 0.075$), so the classic one-way ANOVA is the primary test. It gives $F = 2.80$ with 2 and 36 degrees of freedom, $p = 0.074$ and $\eta^2 = 0.13$: there is only weak evidence that large counties grew more slowly. The Welch ANOVA ($p = 0.032$) and the Kruskal–Wallis test ($p = 0.045$) are reported as sensitivity checks, and no Tukey pair is significant. Figure 19 shows the group distributions and Table 10 collects the tests.

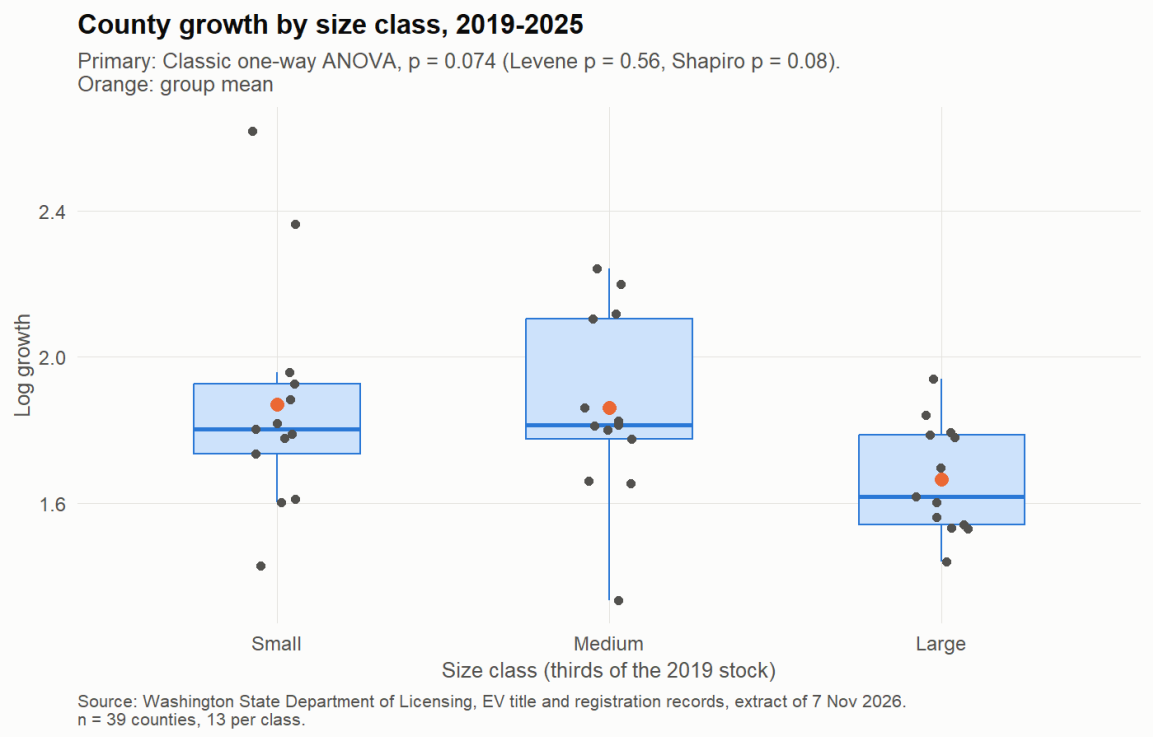


Figure 19: County growth, 2019–2025, by county size class

Table 10: One-way ANOVA: primary test and sensitivity checks

Test	Statistic	df1	df2	p	Role
Classic one-way ANOVA	2.80	2	36.0	0.074	Primary (chosen by the assumption checks)
Welch ANOVA (unequal variances)	4.05	2	21.8	0.032	Sensitivity check
Kruskal–Wallis (no normality)	6.22	2		0.045	Sensitivity check

Source: `outputs/tables/13_anova_oneway_primary.csv`, *section 13.10. Levene* $p = 0.559$*; Shapiro–Wilk* $p = 0.075$*;* $\eta^2 = 0.13$.

A two-way ANOVA adds the side of the Cascade crest. Size and side are strongly confounded (11 of the 13 large counties are western), so with sequential sums of squares the two main effects

depend on the order in which they enter the model and cannot be separated. Only the interaction is stable ($F = 4.57$, $p = 0.018$): the size effect differs between eastern and western Washington (Figure 20). The cells are small (2 to 11 counties), so the interaction should be read with care rather than as a stable regional law.

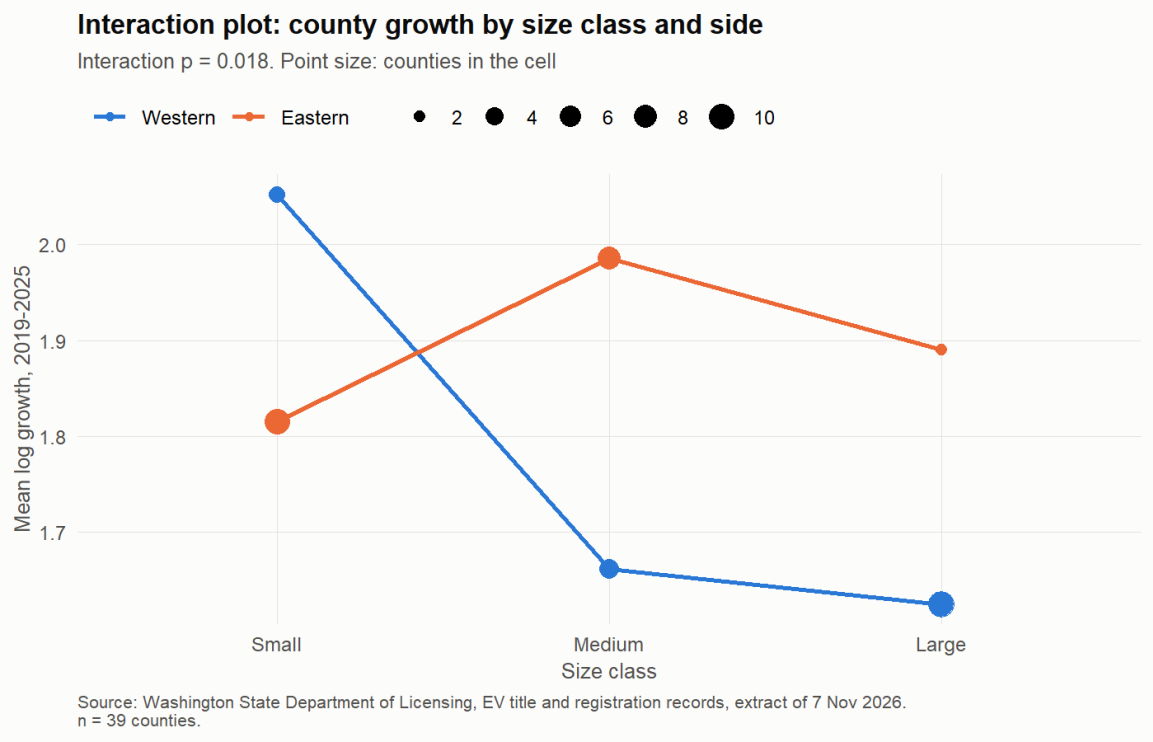


Figure 20: Two-way ANOVA: county size class by side of the Cascades

The same counties are then followed over time. A repeated-measures ANOVA on county BEV share for 2019, 2022 and 2025 (35 counties with 10+ vehicles in 2019; unweighted county means) shows a rise from 56% to 74% (Figure 21). Mauchly's test rejects sphericity ($p < 0.001$), so the Greenhouse–Geisser correction is used: the year effect remains highly significant ($p \approx 10^{-14}$) with partial $\eta^2 = 0.78$, the Friedman test agrees, and all three Bonferroni pairwise comparisons are significant. A mixed ANOVA adds the side of the state as a between-county factor: the side-by-year interaction is $F = 7.02$ ($p = 0.002$), because eastern counties started lower (51% against 61% in 2019) and caught up (73% against 75% in 2025), as Figure 22 shows. The assumption checks are in section 13.9 and the ANOVA models in sections 13.10–13.13 of the R script.

3.4 Regression

The simplest regression asks whether counties that started with a larger stock grew more slowly. Regressing log growth from 2019 to 2025 on log 2019 stock (natural logarithms) gives

$$\widehat{\text{growth}} = 1.990 - 0.038 \, \log(\text{stock}_{2019}),$$

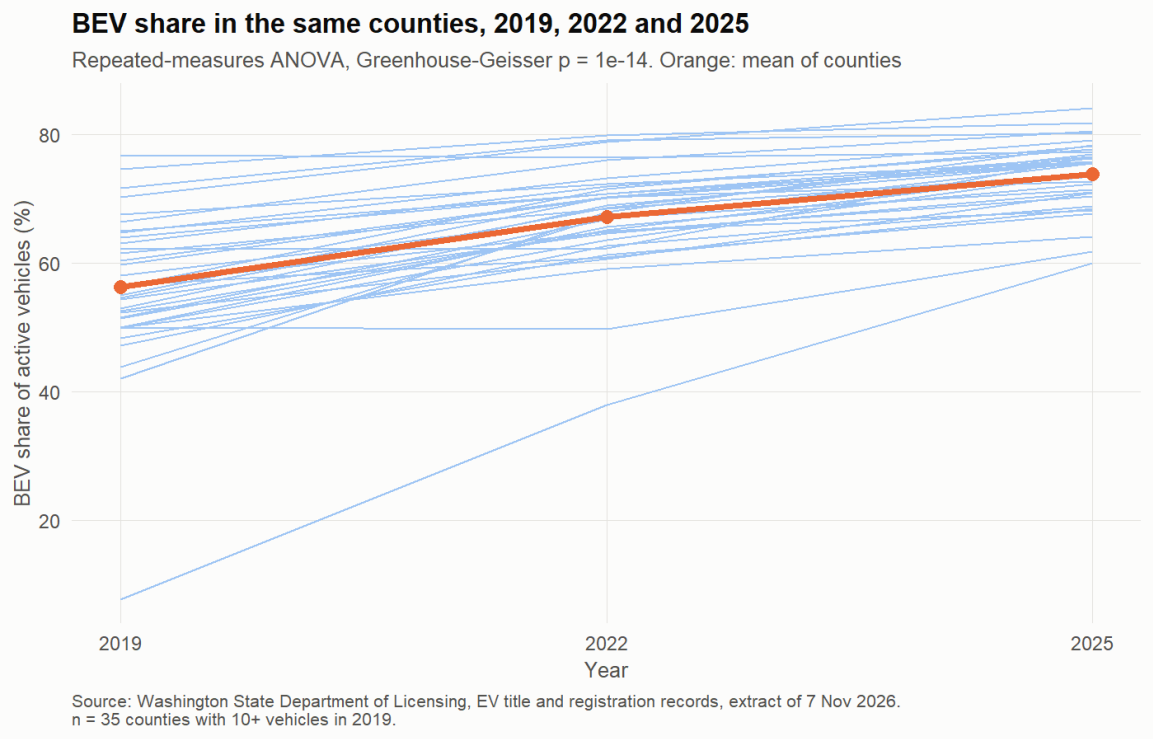


Figure 21: BEV share in the same counties in 2019, 2022 and 2025

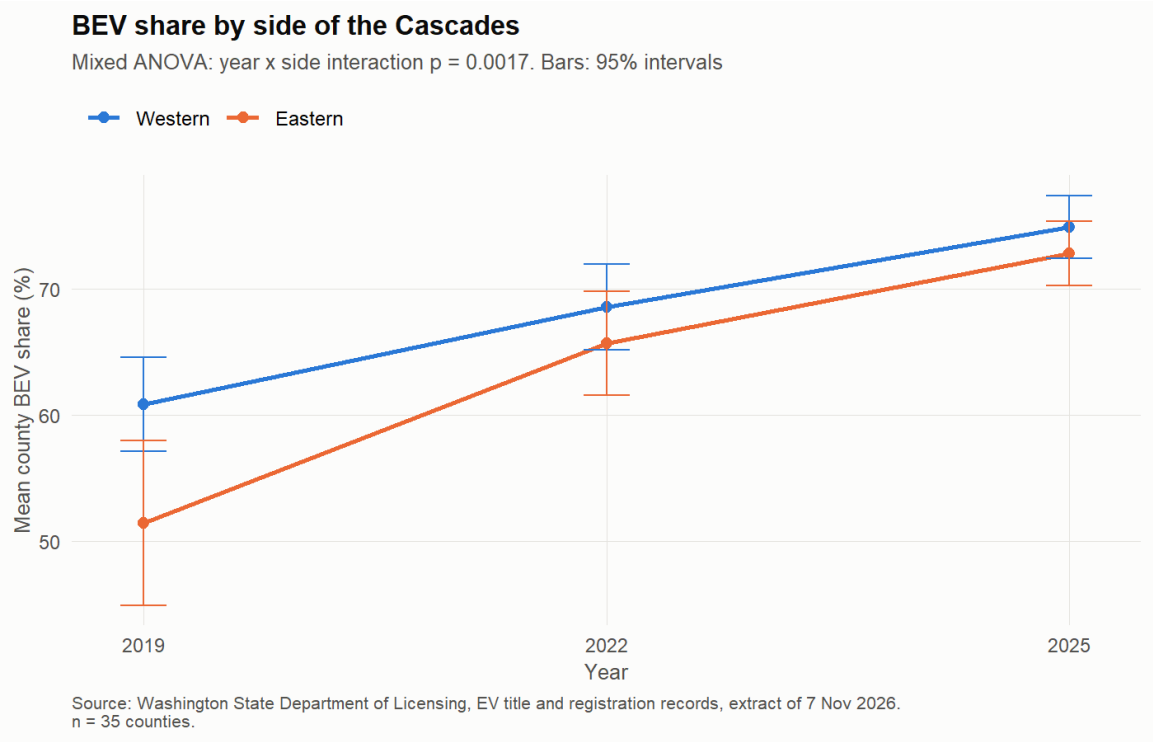


Figure 22: BEV share by side of the Cascades (mixed ANOVA)

with a slope 95% CI of -0.076 to 0.001, $p = 0.058$ and $R^2 = 0.09$: the relationship is in the expected direction but explains little of the variation across 39 counties. Table 11 turns the fitted model into confidence and prediction intervals at four starting stocks. At a 2019 stock of 1,000 vehicles, the fitted log growth is 1.73 (95% CI 1.62 to 1.84), but the 95% prediction interval for a single county is 1.21 to 2.25. The prediction interval is three to six times wider than the confidence interval: the average county is estimated well, a particular county is not.

Table 11: Confidence and prediction intervals from the simple county regression

2019 stock	Fitted log growth	CI low	CI high	PI low	PI high
10	1.90	1.77	2.04	1.38	2.43
100	1.82	1.73	1.90	1.30	2.33
1,000	1.73	1.62	1.84	1.21	2.25
10,000	1.64	1.46	1.82	1.10	2.18

Source: `outputs/tables/14_county_intervals.csv`*, section 14.1. CI = confidence interval; PI = prediction interval.*

A multiple regression adds the share of the 2019 stock that did not enter as new and the 2019 BEV share (both as proportions):

$$\widehat{\text{growth}} = 1.633 - 0.040\ \log(\text{stock}_{2019}) + 1.108\,\text{used}_{2019} + 0.154\,\text{BEV}_{2019}.$$

It raises R^2 to 0.26 (adjusted 0.19), but with HC3 robust standard errors no predictor is significant at 5% (used share +1.11, $p = 0.089$); all variance inflation factors are at most 1.47, so collinearity is not a problem. The diagnostics of the simple regression in Figures 23–25 show no severe departure from the linear model. The four counties above the Cook's distance cut-off are the smallest ones—Wahkiakum, Garfield, Columbia and Adams, with 1 to 11 vehicles in 2019—so their growth rates rest on a handful of vehicles. The models are fitted in sections 14.1–14.2 of the R script.

A separate specification model explains the real price of used vehicles. Regressing log real price on age, mileage, type and make gives $R^2 = 0.67$ on 141,153 sales: each additional year of age is associated with an 11.4% lower real price, each 10,000 additional miles with a 4.7% lower price, and PHEVs sell for 8.1% more than BEVs when the other fields are held fixed. A related hedonic model with sale-year terms gives the used-price index in Section 3.5 of this report. The specification model's residuals are shown in the output file `14_04_spec_residuals.png`.

The logistic regression asks a different question: given what is known at the end of a year, how well can we predict whether a vehicle will have no record in the next two years? Model A uses the base predictors (age, type, make, region, use class and prior transfers); model B adds five features from the vehicle's own history: missed years, late renewals, county moves, the type of the last record and the quarter of the last record. Both are trained on the 2021 baseline and tested on 2023, so the test outcome is not used in fitting. Discrimination is weak: the test AUC

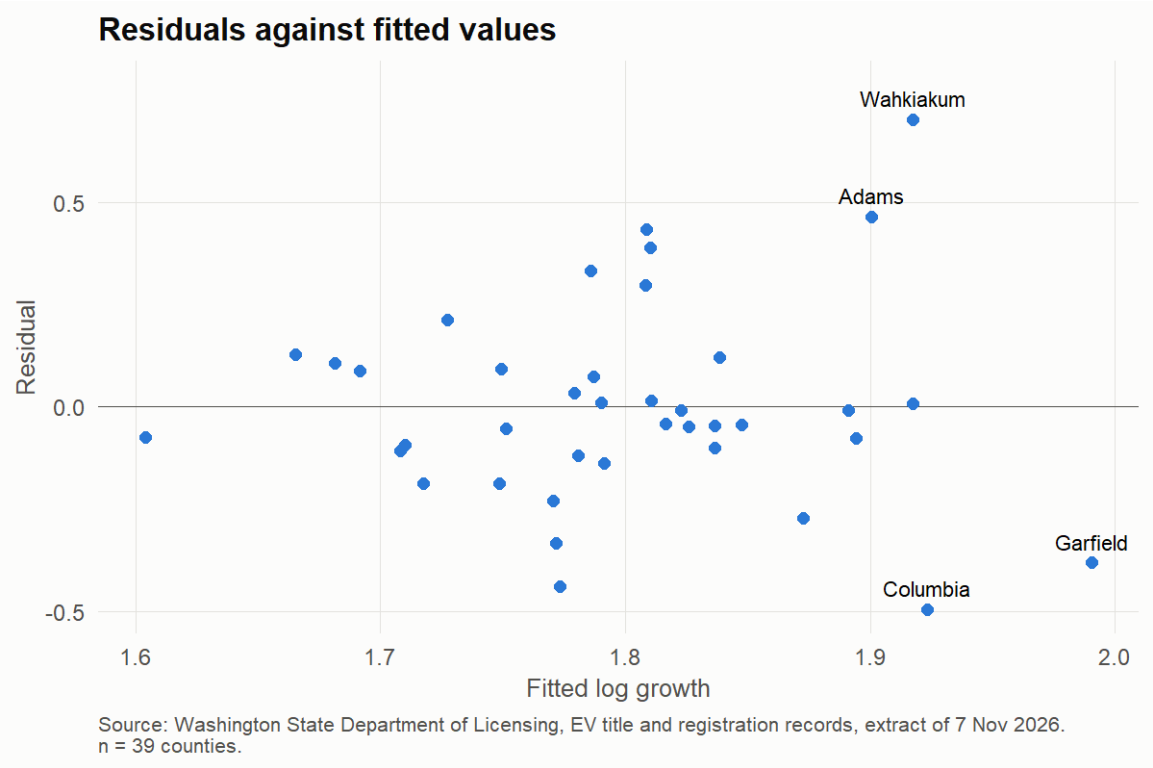


Figure 23: Residuals against fitted values for the simple county regression

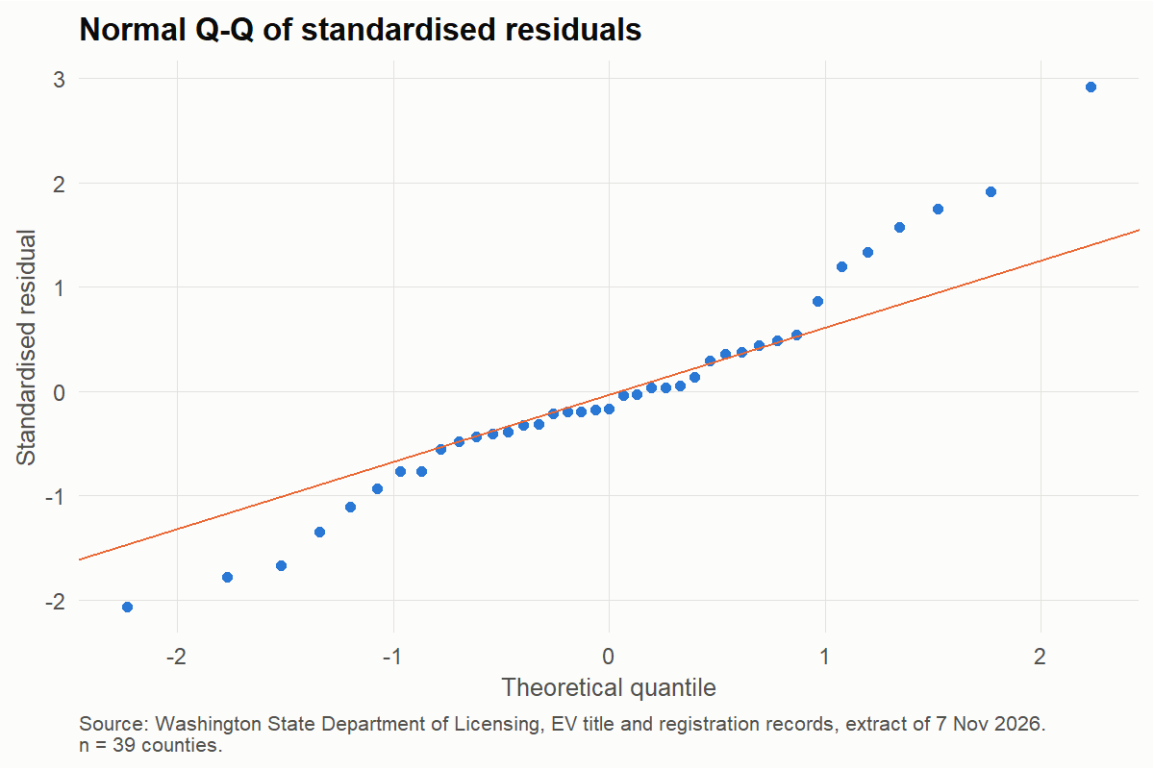


Figure 24: Normal Q–Q plot of standardised residuals for the simple county regression

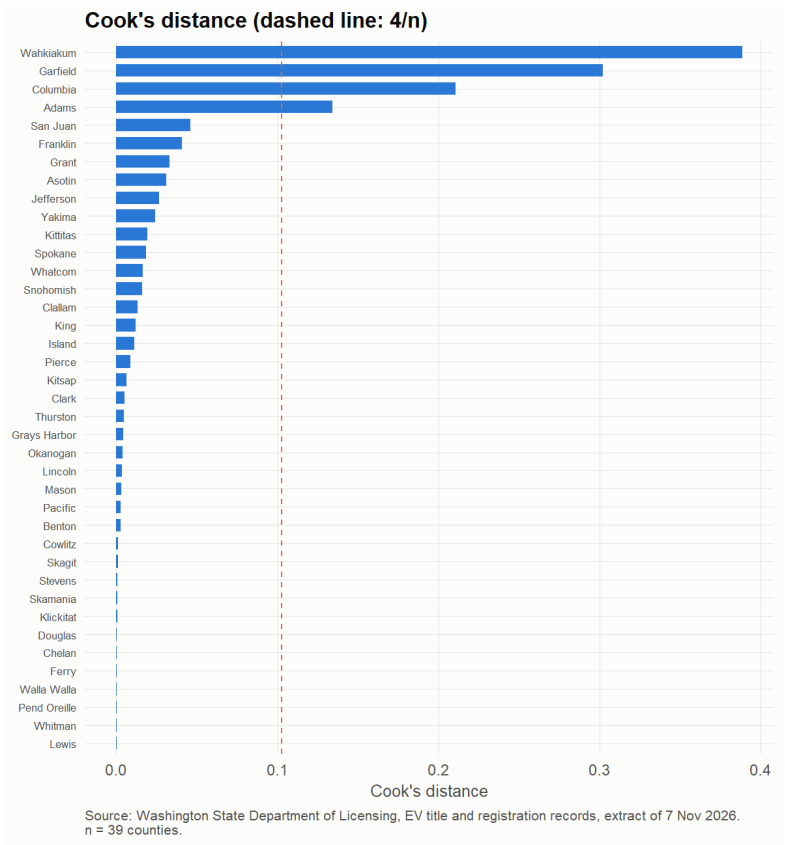


Figure 25: Cook's distance for the simple county regression (dashed line: $4/n$)

is 0.522 for model A and 0.569 for model B, the improvement is statistically detectable (DeLong $p \approx 10^{-76}$), and McFadden's R^2 is 0.007 and 0.044 respectively (Table 12). Figure 26 shows the ROC curves and Figure 27 the calibration by predicted-probability tenth.

Table 12: Logistic models of two-year exit: fit and discrimination

Model	AIC	McFadden R^2	AUC train	AUC test	Brier (test)	Youden cut-off
A: base predictors	44,987	0.007	0.563	0.522	0.070	0.075
B: plus history	43,318	0.044	0.612	0.569	0.067	0.077

Source: `outputs/tables/14_logistic_performance.csv`*, sections 14.4–14.7. Model B, with the lower training AIC, is the final model; model A gives the interpretable odds ratios. Model B adds five history features (missed years, late renewals, county moves, last-record type and quarter).*

Calibration is more informative than discrimination here. Both models reproduce the overall exit rate (mean predicted 7.7% and 7.6% against 7.5% observed), and model B's tenths run from 5.6% to 13.4% observed exit and follow the diagonal in the middle, although it over-predicts the top tenth (15.5% predicted against 13.4% observed). Model A's observed rates span only 6.4% to 8.4%, so it barely separates leavers from stayers. The adjusted probabilities make the associations concrete: model A predicts a 6.9% two-year exit in King County against about 8% elsewhere, 7.1% with no prior transfer against 10.8% with two or more, and 15.2% for truck or commercial use against 7.4% for passenger use; PHEVs are at 6.5% against 7.9% for BEVs after

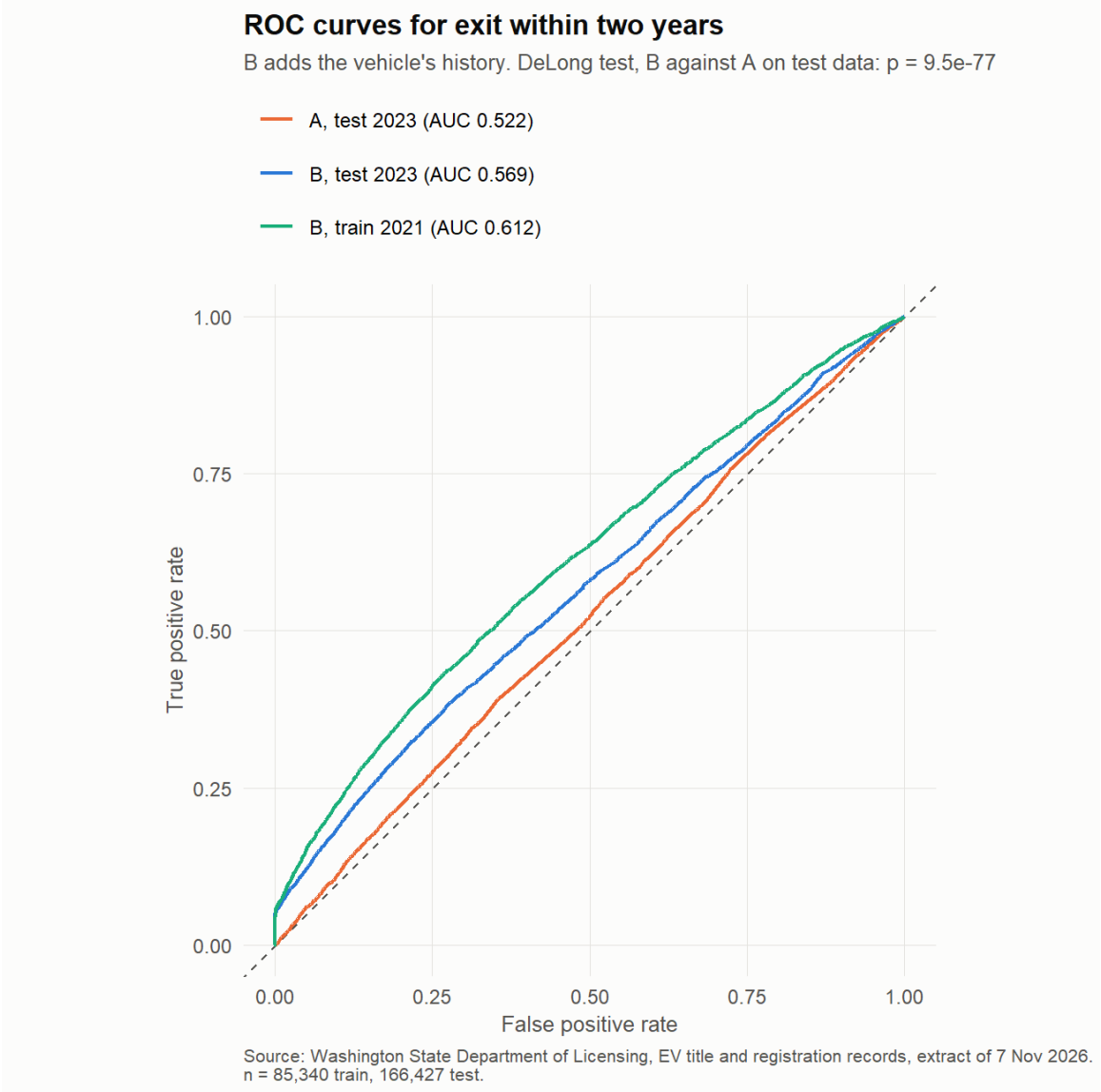


Figure 26: ROC curves for the exit model on the 2023 test data

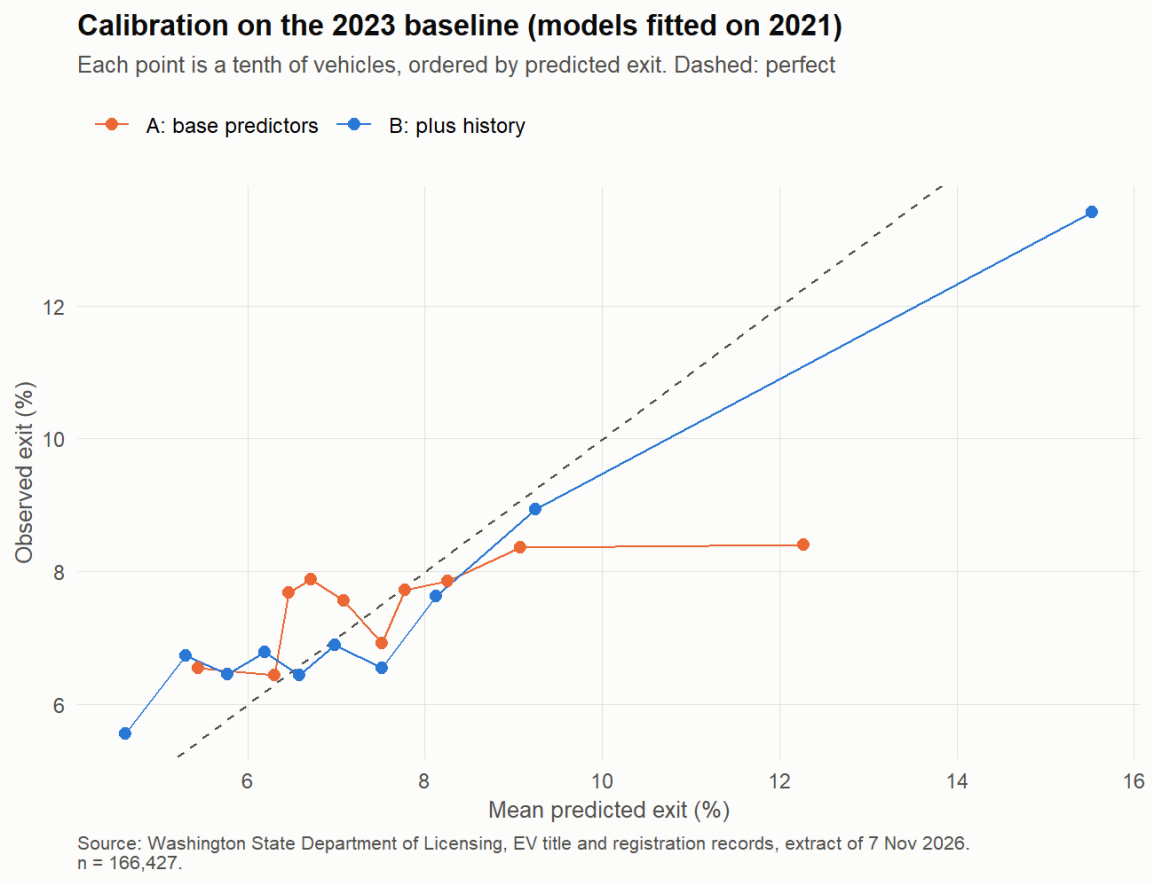


Figure 27: Calibration on the 2023 test data: predicted against observed exit, by tenth

adjusting for make and the other predictors (odds ratio 0.81), a result that differs from the age-only test. These associations come from the 2021 baseline; in the 2019 sensitivity fit the transfer and region odds ratios are close to 1. Removing leavers that later returned barely changes the training AUC (0.563 to 0.555 for model A, 0.612 to 0.606 for model B), so label noise is not the reason the models are weak. At a 0.5 cut-off almost no vehicle is predicted to leave, because only 7.5% leave, and at the Youden cut-off model B has sensitivity 0.40 and specificity 0.70. Figure 28 shows the decision boundary of a two-predictor model on age and tenure, drawn at the 7.5% training exit rate rather than at 0.5, and Figure 29 the predicted probabilities of leavers and stayers.

The linearity of the logit was checked with Box–Tidwell terms before the final model was fixed. Age passes the check ($p = 0.79$) and enters as a straight-line term, whereas tenure shows curvature ($p < 0.001$) and is used only in the two-predictor decision boundary, where this is stated as a limitation. The regression and model checks are implemented in sections 14.1–14.9 of the R script.

3.5 Extensions beyond the brief

Seven extensions deepen the answer without changing its direction. They are ordered below by section number (which differs from their extension number, because they build on one another).

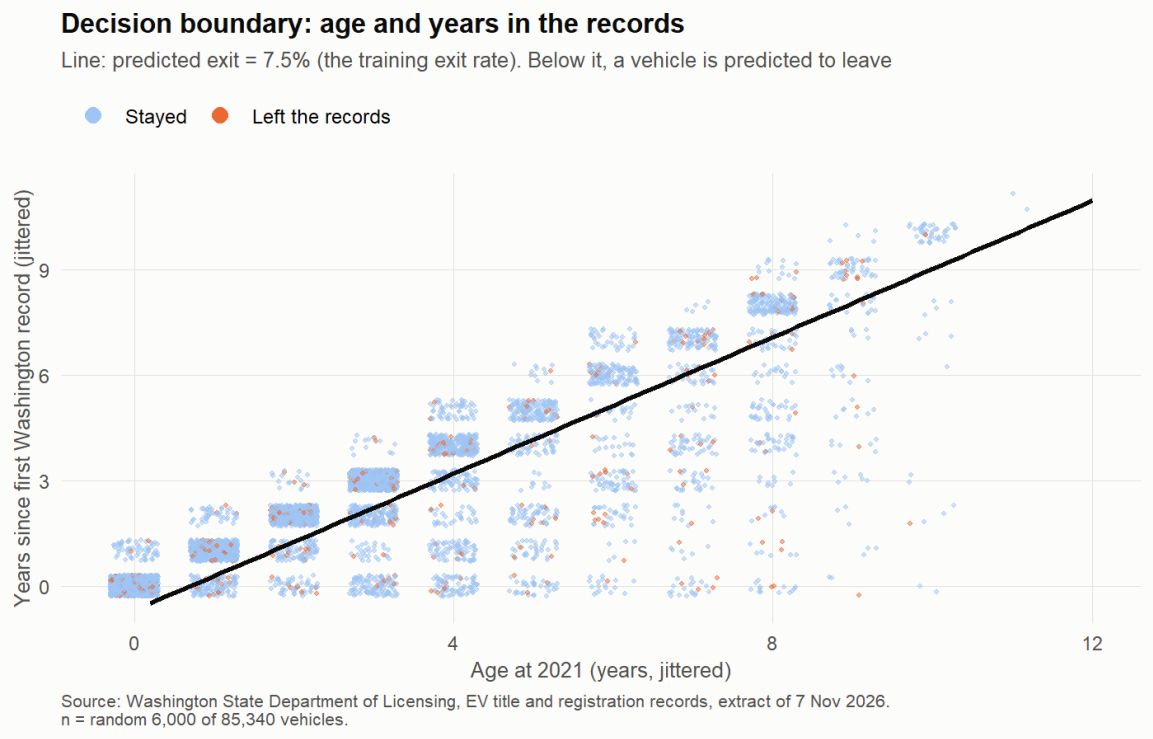


Figure 28: Decision boundary of a two-predictor logistic model (age and years in the records): the line is a predicted exit of 7.5%, the training exit rate

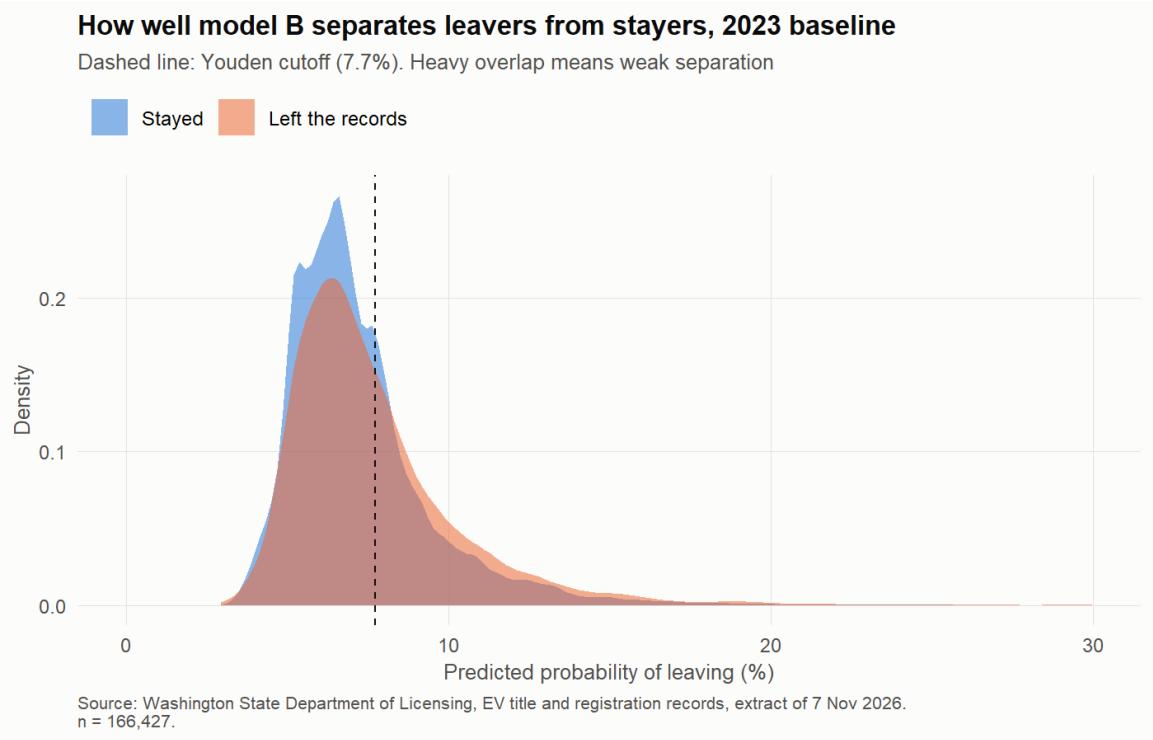


Figure 29: Predicted probabilities for vehicles that left and stayed in the 2023 test data

Extension 1 (section 15) builds a used-price index from repeat sales. There are 20,631 repeat-sale pairs on 17,088 cars. A repeat-sales index includes the effect of ageing, while a hedonic index holds age and mileage fixed; the hedonic index rises by 21% in 2022 and falls by about 17% in each of 2023 and 2024 (Figure 30), and each year of age is associated with an 11.4% lower real price.

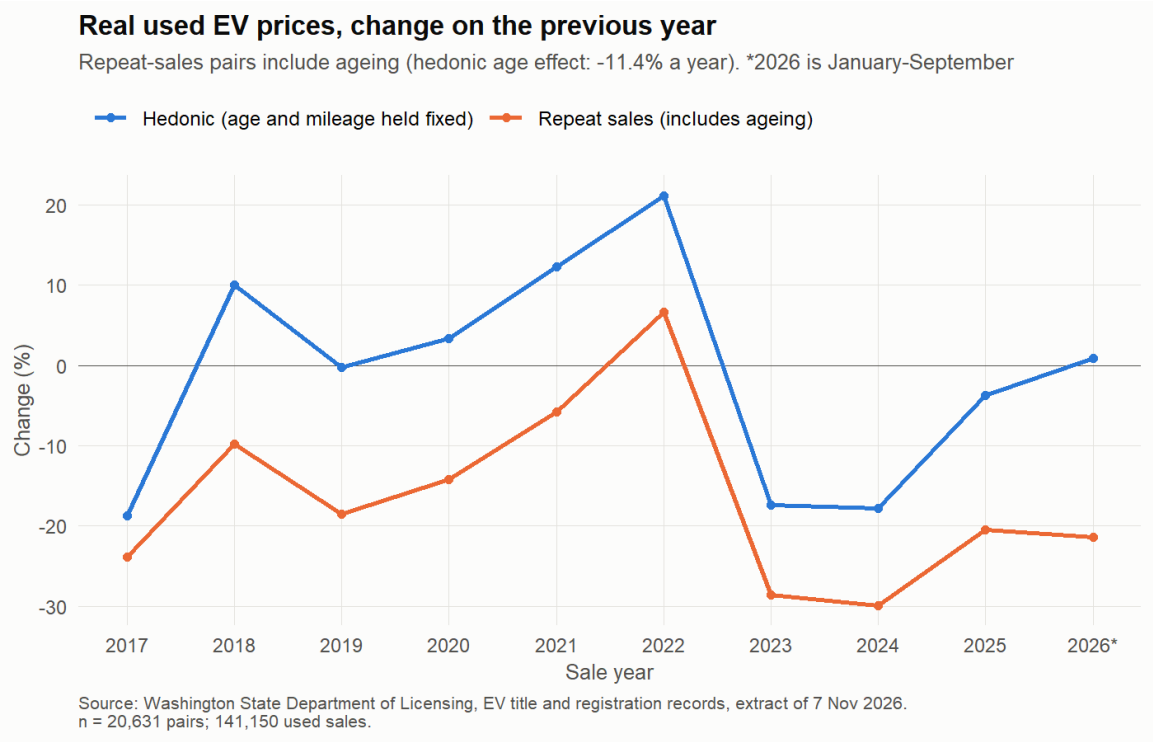


Figure 30: Real used electric-vehicle prices, change on the previous year

Extension 3 (section 16) examines county movement. In a year with a title transfer, 37.1% of King County vehicles left the county, against 2.1% in a year without a transfer (Figure 31); a title transfer is therefore strongly associated with a move, although the records still cannot say where the vehicle went.

Extension 4 (section 17) follows cohorts by age. At age 5, 72% of BEVs and 74% of PHEVs are still in the records; at age 10 the shares are 53% and 55% (Figure 32). A Cox model gives a PHEV hazard ratio of 0.92 over 2010–2023, but the ratio is 0.69–0.79 up to 2019 and 1.04 and 1.03 for 2019–2021 and 2021–2023, and a Schoenfeld residual test rejects the proportional-hazards assumption ($p = 0.024$): the PHEV effect is not constant, so 0.92 is an average over periods. The cohort evidence is therefore consistent with the main test: the difference in favour of PHEVs belongs to the earlier period.

Extension 5 (section 18) tests whether electric range predicts leaving. Among BEVs of model year 2020 or earlier, the odds ratio is 0.94 per 100 miles ($p = 0.38$), so there is no evidence that range is associated with leaving in this population.

Extension 2 (section 19) places the counts on a per-resident basis using Census county popu-

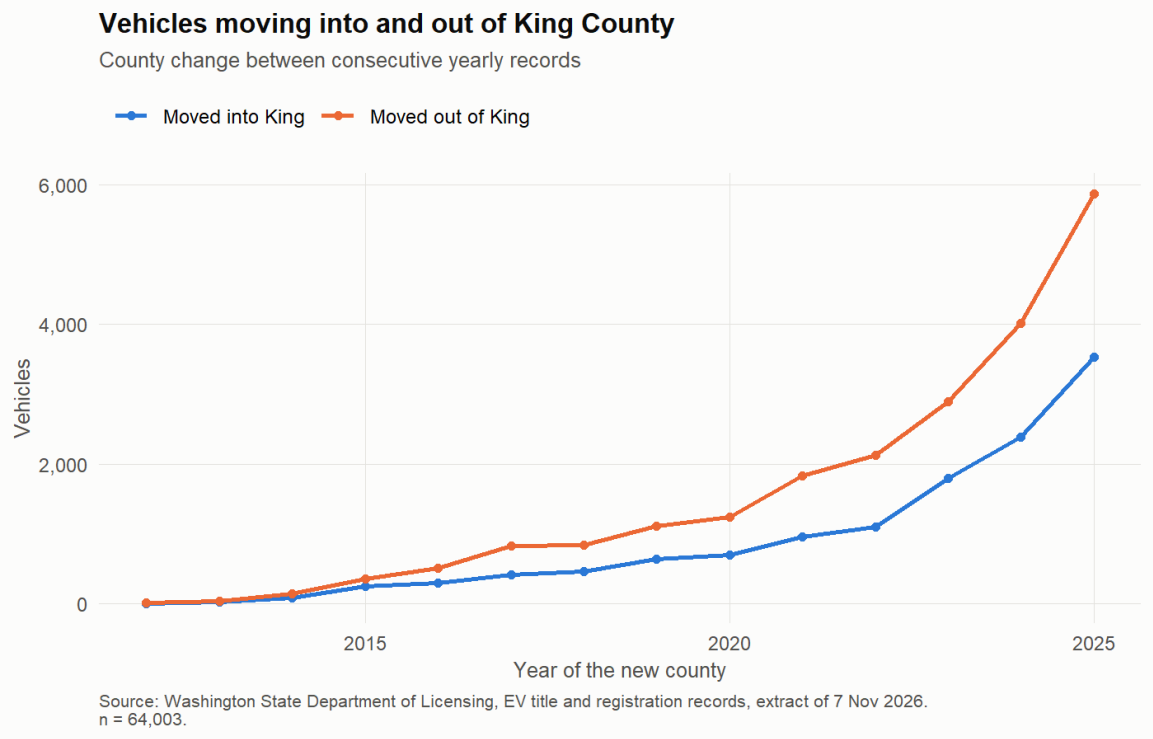


Figure 31: Vehicles moving into and out of King County

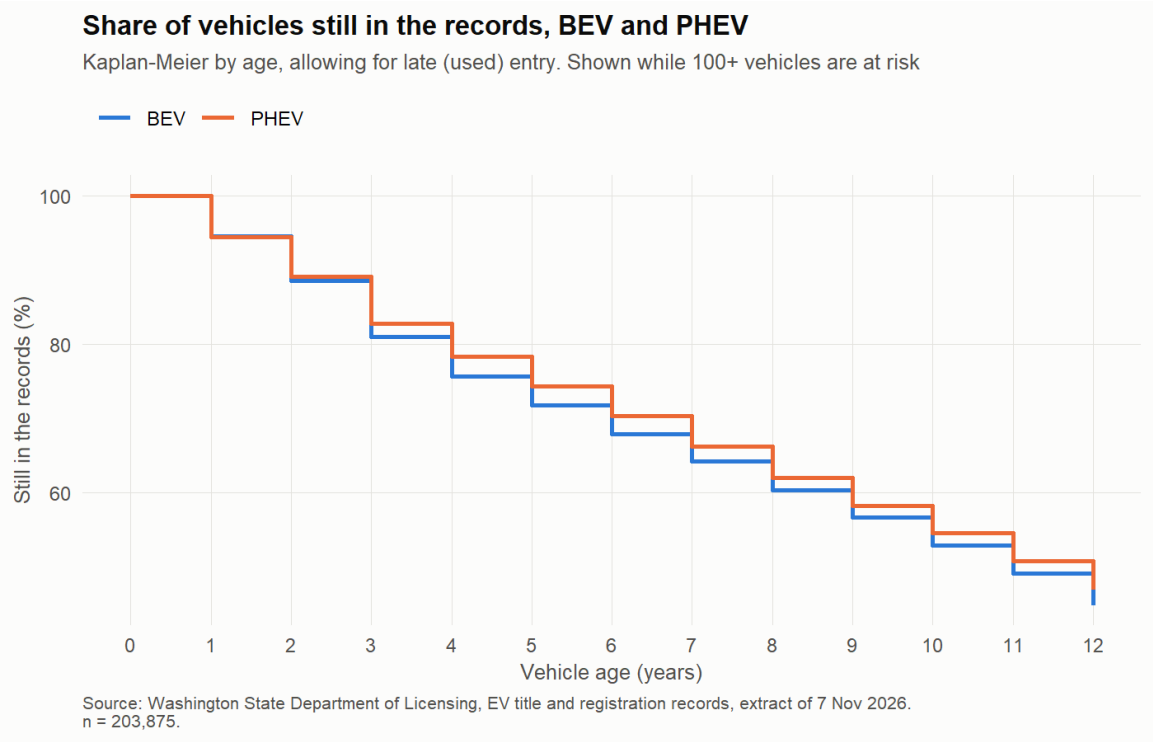


Figure 32: Share of vehicles still in the records by age, BEVs and PHEVs

lation. In 2025 the state had 33.8 vehicles per 1,000 residents, King County 57.2 and the rest of the state 24.0; San Juan County had the highest rate (66.2). King County holds 49.7% of vehicles but only 29.3% of residents (Figures 33 and 34).

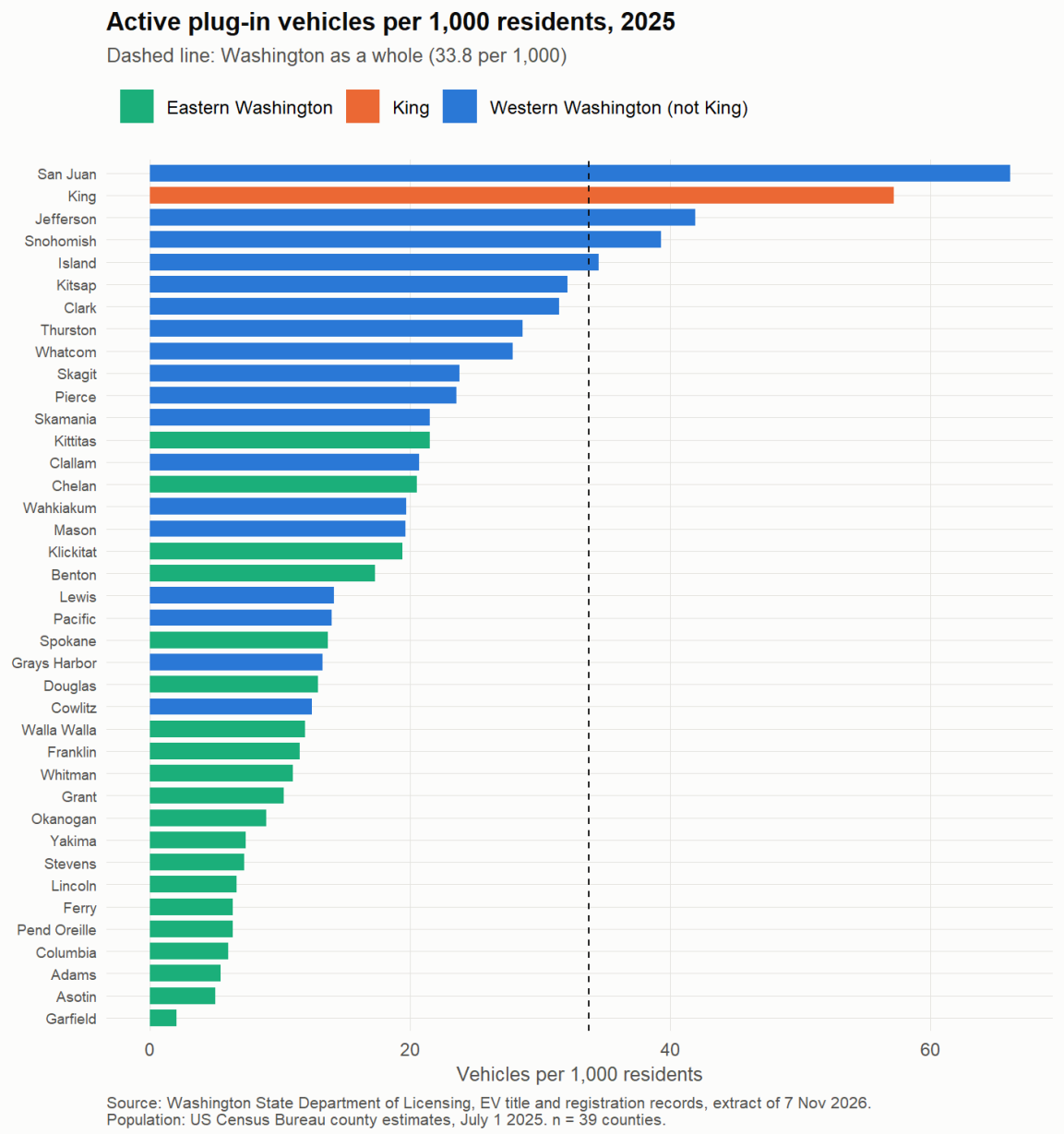


Figure 33: Plug-in vehicles per 1,000 residents by county, 2025

Extension 6 (section 20) projects the population to 2028 with the stock-and-flow identity, on the gap-filled count (286,478 in 2025). Entries stay at their 2025 level in the primary (Level) scenario, with the exit rate, return rate and entry volatility estimated from the recent past and propagated through 10,000 simulated paths. The central projection is about 444,000 active vehicles in 2028, with a 95% interval of 331,000 to 714,000; the Trend scenario, in which entries keep their average growth, gives about 593,000 (Figure 35). A backtest that cuts the data at 2022 is within 1% of the actual 2023–2025 counts with actual entries; with projected entries, Level was 30% low and Trend 2% high in 2025, because entries jumped in 2023, and both 80%

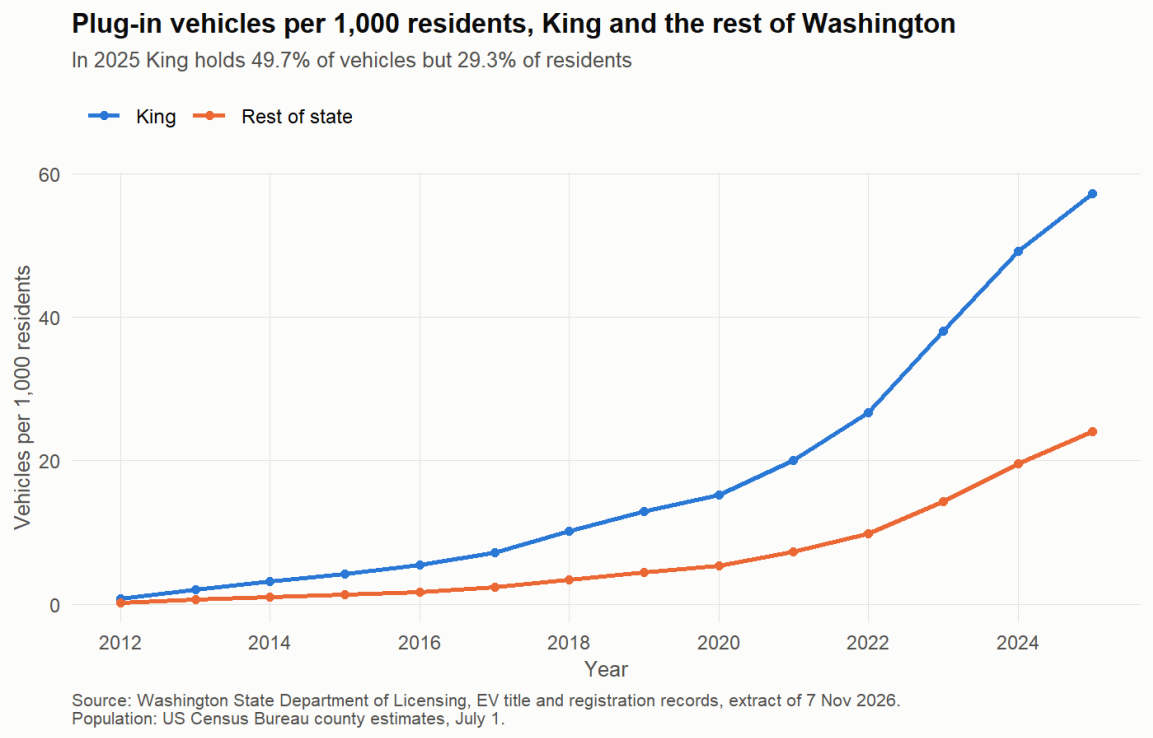


Figure 34: Vehicles per 1,000 residents, King County and the rest of the state

intervals contain the actual counts. Level is kept as the primary scenario because entries were flat in 2024–2025 and the state sales-tax exemption ended on 31 July 2025. January–September 2026 entries are 11.8% below the same months of 2025, implying about 66,000 for the year, below Level's 72,000. The projection assumes constant rates, and it has only one backtest origin, so it should be updated with later extracts.

Extension 7 (section 21) looks at legislative districts and utility territories, the geography that planners actually use. The five largest districts held 31% of vehicles in 2019 and 26% in 2025; districts 41, 45 and 48 have about 89–102 vehicles per 1,000 residents (assuming equal district populations), while district 15 has about 5 (Figure 36). The three largest territory labels (Puget Sound Energy, Seattle and Tacoma) hold 73% of vehicles, so charging demand and grid planning are concentrated in a small number of service areas. The district and utility analyses are in sections 21.1–21.4 of the R script.

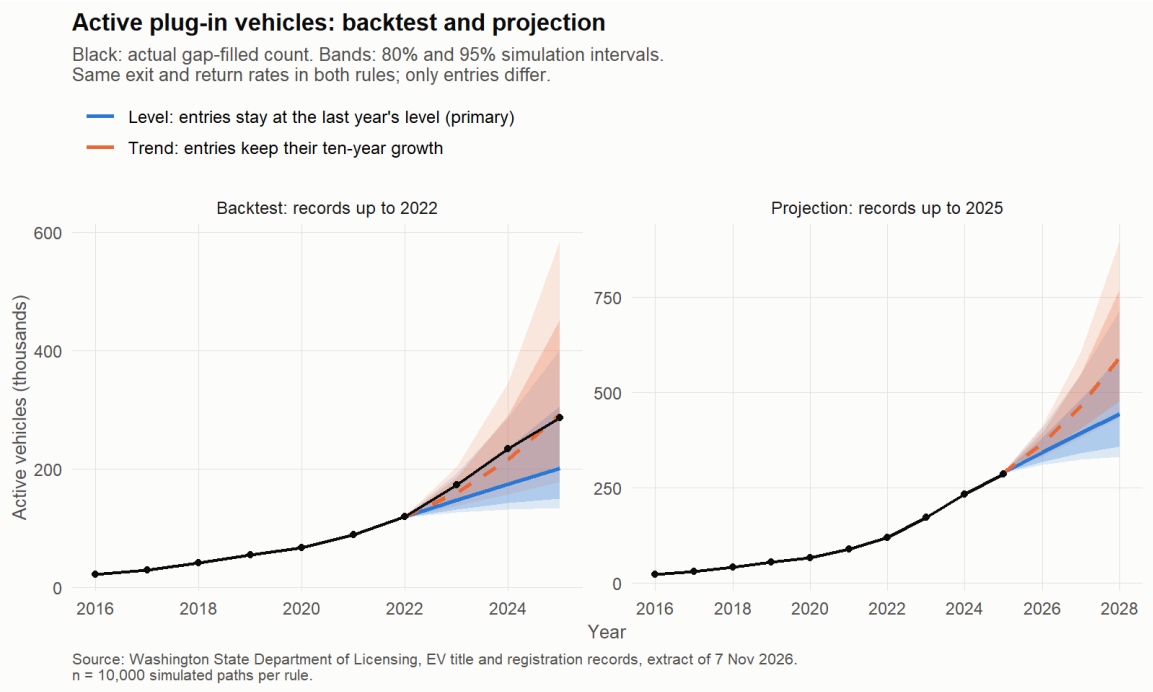


Figure 35: Active vehicles projected to 2028, with 80% and 95% intervals

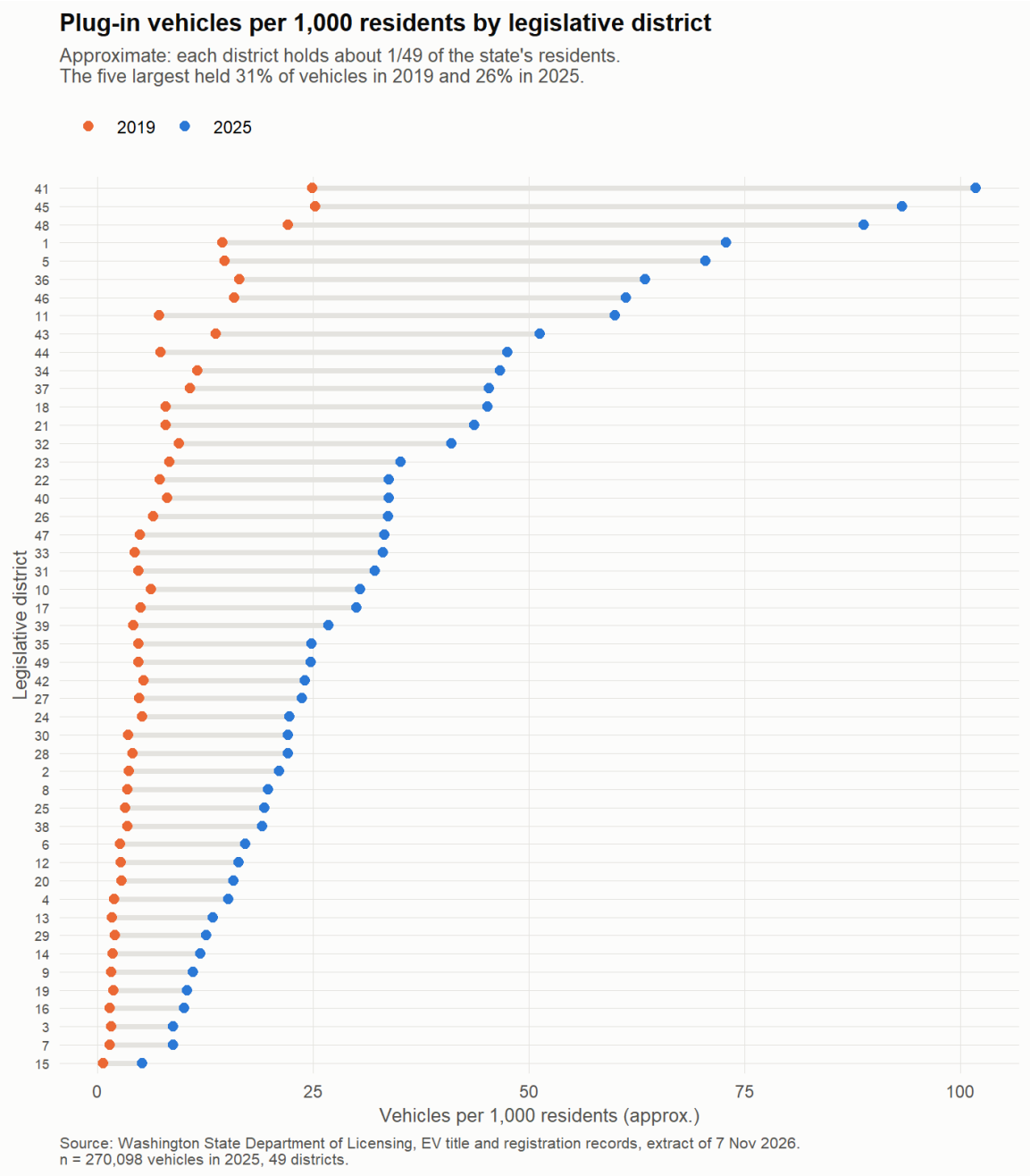


Figure 36: Vehicles per 1,000 residents by legislative district

4. Actionable Conclusion and Insight

4.1 Key findings

The population grew about five-fold between 2019 and 2025, from 52,710 to 270,098 active vehicles, and it aged at the same time: mean age rose from 2.58 to 3.05 years. The gap-filled count (54,733 in 2019 and 286,478 in 2025) shows that the growth is not an artefact of missing records, and the stock-and-flow identity and benchmark checks make the counts trustworthy.

The fleet also spread out. King County's share fell 5.5 percentage points, eastern counties caught up on BEV share, and the share held by the five largest legislative districts fell from 31% to 26%. King County still has 49.7% of the vehicles and 29.3% of the residents, so the concentration remains substantial even as it declines.

Which vehicles stay is now clearer. In the recent fleet, no significant difference between BEVs and PHEVs was detected after adjusting for age (odds ratio 1.03, 95% CI 0.97–1.10); with make and the other predictors also held fixed, PHEVs leave somewhat less often (odds ratio 0.81). The clearer difference in favour of PHEVs is in the 2019 and earlier baselines, and range shows no association with leaving. In the 2021 baseline, leaving is associated with prior transfers, use class and region, but it is hard to predict (test AUC 0.57), because the reasons a vehicle leaves—moving out of state, an accident, a sale to another state—are not in the records.

About 7–7.5% of active vehicles leave the records each year, with no record in the following two years (6.9% from the 2019 baseline and 7.5% from 2023). This is the attrition rate that a stock-and-flow forecast should use, and it is low enough that entries, not exits, dominate the growth of the fleet.

4.2 Practical value

Planners and utilities. Charging and grid demand is concentrated in a few districts and three utility territories, but growth is now faster outside King County, so new capacity should be planned there as well. The projection gives a range for load planning—about 444,000 vehicles in 2028, with a wide interval—and should be updated each year, because the 2026 check shows that entries can shift quickly. Per-resident rates identify under-served areas: district 15 has about 5 vehicles per 1,000 residents against 102 in district 41.

State revenue and fleets. Electric vehicle registration fees depend on the active stock, not on sales, and the stock-and-flow method (entries, an annual exit rate of about 7%, and returns) is a transparent way to forecast that base. The end of the sales-tax exemption in July 2025 coincides with 11.8% lower entries in January–September 2026, so revenue planning should not assume that the 2023–2025 entry level continues. For fleet managers, BEVs and PHEVs show no clear difference in retention once age is accounted for, and used prices fall about 11% per year of age, which is useful for replacement and resale planning.

Policy. Equity should be monitored, because rural and eastern areas still have far fewer vehicles per resident. The records cannot say why a vehicle leaves; recording out-of-state transfers separately from vehicles taken off the road would make retention measurable and would strengthen the models considerably. Population counts should be published with the rules used (duplicates, exempt classes, two-year exit rule), because simple county sums double-count vehicles that move.

4.3 Limits and future research

The results are associations, not causes, and leaving is inferred from missing records. Exits are observable only to 2023, the 2025 gap-filled count and the 2024 leaver count are provisional, and 2026 is partial. The projection holds rates constant and has a single backtest origin. Future work could link the records with public charging-station data, out-of-state title transfers and owner characteristics, none of which are in this dataset. The assignment brief and marking rubric are reproduced at the end of this report.

APPENDIX 1

MARKING RUBRICS

Component Title		Assignment				Percentage (%)	100	
Criteria		Score and Descriptors					Weight (%)	Marks
		Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
	Data Pre-processing & Feature Engineering	Exceptional and innovative feature extraction (e.g., Phase Coherence). Robust and clearly justified cleaning of raw hardware errors.	Good feature extraction. Raw data cleaning is mostly effective with minor oversights.	Basic feature extraction. Minimal data cleaning applied.	Feature extraction is unclear or poorly executed. Cleaning is ignored.	No meaningful feature extraction or cleaning. Relies on uncleaned data.	**15**	
	Descriptive Stats & Correlation	Comprehensive analysis of central tendency, variability, and symmetry. Insightful interpretation of correlation.	Most statistics calculated correctly with good interpretation.	Appropriate statistics applied but basic interpretation.	Some statistics applied but poorly interpreted.	Statistics missing or incorrectly applied.	**10**	
	Probability & Distributions	Deep, highly accurate modeling of packet loss/arrival. Excellent simulation of the Central Limit Theorem.	Probability concepts applied clearly with minor calculation or plotting issues.	Basic application of probability models. Simulation has some flaws.	Models are poorly applied or misinterpreted. Simulation is incorrect.	Probability and distributions are skipped or completely wrong.	**15**	
	Visualizations	Visuals beautifully highlight temporal changes and class separation. Flawlessly exported as PNGs with perfect labeling.	Most visualizations are clear and appropriately labeled.	Appropriate visuals present but lack clarity or proper labeling.	Visuals are unclear, mislabeled, or poorly integrated.	Visuals are missing, irrelevant, or highly confusing.	**10**	
	Hypothesis Testing	Tests performed accurately to validate feature separation. Insightful interpretation of p-values and assumptions.	Most tests performed correctly with good interpretation.	Appropriate tests performed but interpretation is basic.	Tests applied to the wrong variables or poorly interpreted.	Hypothesis testing is missing or fundamentally flawed.	**10**	
	Regression Analysis	Logistic Regression model is robust. Deep understanding of decision boundaries and logit linearity checks.	Model constructed well. Clear findings and good interpretation.	Adequate model constructed but lacks depth in evaluating assumptions.	Regression poorly specified or interpretation of odds is incorrect.	Regression model missing or completely misapplied.	**10**	
	Report Content & Format	Exceptionally well-structured, logical flow. Flawlessly	Well-organized with clear explanations. Minor	Covers basic requirements but lacks depth. Several	Lacks structure, missing key sections, and	Highly disorganized, missing critical content.	**10**	

		adheres to formatting rules.	formatting issues.	formatting inconsistencies.	poor formatting.			
	Content Delivery (Presentation)	Engaging, professional, clear, and confident delivery. Demonstrates deep mastery of the concepts.	Clear and well-organized presentation. Good understanding.	Generally clear but lacks organization or pacing.	Unclear and disorganized. Weak understanding of the analysis.	Poor or no presentation.	**10**	
	Technical Q&A	Excellent knowledge. Easily answers complex questions regarding R code, data cleaning, and model logic.	Good knowledge. Able to answer most questions with solid reasoning.	Adequate knowledge. Struggles with deeper technical questions.	Limited knowledge. Cannot explain the rationale behind choices.	Unable to answer basic questions about the project.	**10**	
						TOTAL	**100**	